

ELISA analysis

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First time run order: Spleen → CBC → ELISA

Requires clean__meta__data.csv from the spleen analysis file.

Loading the libraries

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.1
## v lubridate  1.9.4      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readxl)
library(writexl)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##      chisq.test, fisher.test
```

```
library(factoextra)
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```
library(skimr)
library(naniar)
```

```
##
## Attaching package: 'naniar'
##
## The following object is masked from 'package:skimr':
##
##     n_complete
```

```
library(ggplot2)
library(ggpubr)
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```
library(lmerTest)
```

```
##
## Attaching package: 'lmerTest'
##
## The following object is masked from 'package:lme4':
##
##     lmer
##
## The following object is masked from 'package:stats':
##
##     step
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
```

```
## The following object is masked from 'package:readr':  
##  
##      col_factor
```

```
library(haven)  
library(broom.mixed)  
library(readr)  
library(here)
```

```
## here() starts at /Users/zehao/Documents/Research Projects/Graham Group Thesis Research (Summer 2025)
```

```
library(DHARMa)
```

```
## This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package = 'DHARMa')
```

```
library(rstatix)
```

```
##  
## Attaching package: 'rstatix'  
##  
## The following object is masked from 'package:janitor':  
##  
##      make_clean_names  
##  
## The following object is masked from 'package:stats':  
##  
##      filter
```

Setting the paths

```
base_path <- "/Users/zehao/Documents/Research Projects/Graham Group Thesis Research (Summer 2025)"  
  
data_path <- file.path(base_path, "Metadata", "elisadata")  
  
clean_data_path <- file.path(base_path, "Cleaned data")  
  
figure_path      <- file.path(base_path, "Figures")  
elisa_figure_path <- file.path(figure_path, "ELISA Figures")  
dir.create(elisa_figure_path, showWarnings = FALSE, recursive = TRUE)
```

Setting up the functions for repeated codes

Mean calculator

```

mean_calculator <- function(input_data,
                              final_concentration,
                              cv_data,
                              groupings = c("mouse_id",
                                              "treatment",
                                              "plate",
                                              "strain",
                                              "t_muris_dose",
                                              "lab_or_rw",
                                              "cohort",
                                              "experiment",
                                              "source",
                                              "experiment_day")) {
  input_data %>%
    group_by(across(all_of(groupings))) %>%
    summarize(
      mean_concentration = mean({final_concentration}},
                        na.rm = TRUE),
      cv_percent = first(na.omit({cv_data})),
      .groups = "drop"
    )
}

```

Setting the colors (Okabe-Ito palette and Paul Tol's Bright)

```

color_black <- "#000000"
color_mustard <- "#E69F00"
color_light_blue <- "#56B4E9"
color_green <- "#009E73"
color_yellow <- "#F0E442"
color_dark_blue <- "#0072B2"
color_orange <- "#D55E00"
color_purple <- "#CC79A7"

color_grey <- "#BBBBBB"
color_dark_purple <- "#AA3377"
color_light_purple <- "#EE6677"
color_sand <- "#CCBB44"
color_cyan <- "#66CCEE"
color_olive <- "#228833"
color_navy <- "#4477AA"

```

Basic elisa summary plotter

```

elisa_summary_plotter <- function(input_data,
                                   y_label,
                                   title = NULL) {
  ggplot(input_data %>% mutate(treatment = factor(treatment, levels = c("IgG", "LPS", "CD28"))),

```

```

lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")),
t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High"))),
aes(x = lab_or_rw,
y = mean_concentration,
fill = t_muris_dose)) +
geom_boxplot(position = position_dodge(width = 0.75),
alpha = 0.8,
outlier.shape = NA,
width = 0.5,
color = "black",
linewidth = 0.6) +
geom_jitter(position = position_jitterdodge(jitter.width = 0.15,
dodge.width = 0.75),
alpha = 0.8,
size = 1.8,
shape = 21,
color = "black") +
facet_wrap(~ treatment,
labeller = as_labeller(c("CD28" = "CD28SA",
"LPS" = "LPS",
"IgG" = "IgG1")))) +
scale_fill_manual(values = c("Zero" = color_dark_blue,
"High" = color_orange),
name = expression(bolditalic("T. muris") ~ bold("Dose")))) +
scale_color_manual(values = c("Zero" = color_dark_blue,
"High" = color_orange),
name = expression(bolditalic("T. muris") ~ bold("Dose")))) +
scale_x_discrete(labels = c("Lab" = "Laboratory",
"RW" = "Rewilded"),
name = expression(bold("Environment")))) +
scale_y_continuous(expand = expansion(mult = c(0.05, 0.3))) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
color = "black",
linewidth = 0.8),
strip.text = element_text(size = 10,
face = "bold"),
panel.border = element_rect(color = "black",
fill = NA,
linewidth = 0.8),
panel.grid = element_blank(),
axis.title.x = element_text(face = "bold"),
axis.title.y = element_text(face = "bold"),
axis.text = element_text(color = "black"),
axis.ticks = element_line(color = "black",
linewidth = 0.3),
axis.ticks.length = unit(0.15, "cm"),
plot.title = element_text(hjust = 0.5, face = "bold")) +
ylab(y_label) +
ggtitle(title)
}

```

Treatment specific plotter

```
elisa_treatment_plotter <- function(input_data,
                                     y_label,
                                     title = NULL) {
  ggplot(input_data %>% mutate(lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")),
                               t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High"))),
    aes(x = lab_or_rw,
        y = mean_concentration,
        fill = t_muris_dose)) +
  geom_boxplot(alpha = 0.8,
    outlier.shape = NA,
    width = 0.5,
    color = "black",
    linewidth = 0.6,
    position = position_dodge(width = 0.6)) +
  geom_jitter(alpha = 0.8,
    size = 1.8,
    shape = 21,
    color = "black",
    position = position_jitterdodge(jitter.width = 0.1,
                                    dodge.width = 0.6)) +
  scale_fill_manual(values = c("Zero" = color_dark_blue,
                               "High" = color_orange),
    name = expression(bolditalic("T. muris") ~ bold("Dose")) +
  scale_color_manual(values = c("Zero" = color_dark_blue,
                                "High" = color_orange),
    name = expression(bolditalic("T. muris") ~ bold("Dose")) +
  scale_x_discrete(labels = c("Lab" = "Laboratory",
                              "RW" = "Rewilded"),
    name = expression(bold("Environment"))) +
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.3))) +
  theme_minimal() +
  theme(panel.border = element_rect(color = "black",
                                     fill = NA,
                                     linewidth = 0.8),
    panel.grid = element_blank(),
    axis.title.x = element_text(face = "bold"),
    axis.title.y = element_text(face = "bold"),
    axis.text = element_text(color = "black"),
    axis.ticks = element_line(color = "black",
                              linewidth = 0.3),
    axis.ticks.length = unit(0.15, "cm"),
    plot.title = element_text(hjust = 0.5, face = "bold")) +
  ylab(y_label) +
  ggtitle(title)
}
```

Emmeans per treatment data frame maker

NOTE: `emm_treatment` function was removed — it contained a bug where

`filter(treatment == treatment)` compared the column to itself (always TRUE).

The function was never called anywhere, so it was dead code.

Plot saver

```
plot_saver <- function(filename,
                        plot,
                        path = elisa_figure_path,
                        width = 6,
                        height = 4,
                        dpi = 300) {
  ggsave(filename = filename,
          plot = plot,
          path = path,
          width = width,
          height = height,
          units = "in",
          dpi = dpi)
}
```

Loading the data

```
elisa_ifn_1 <- read_xlsx(file.path(data_path,
                                   "IFNg_Plate1_20260121.xlsx"),
                        sheet = "Plate 1 Data")

elisa_ifn_2 <- read_xlsx(file.path(data_path,
                                   "IFNg_Plate2_20260123.xlsx"),
                        sheet = "Plate 2 Data")

elisa_tnf_1 <- read_xlsx(file.path(data_path,
                                   "TNFa_Plate1_20260121.xlsx"),
                        sheet = "Plate 1 Data")

elisa_tnf_2 <- read_xlsx(file.path(data_path,
                                   "TNFa_Plate2_20260123.xlsx"),
                        sheet = "Plate 2 Data")

plate1_map <- read_xlsx(file.path(data_path,
```

```

        "SF ELISA Plate Mapping.xlsx"),
    sheet = "Plate 1 for both IFN and TNF")

plate2_map <- read_xlsx(file.path(data_path,
        "SF ELISA Plate Mapping.xlsx"),
    sheet = "Plate 2 for both IFN and TNF")

meta_data <- read_csv(file.path(clean_data_path,
        "clean_meta_data.csv"))

## Rows: 110 Columns: 7
## -- Column specification -----
## Delimiter: ","
## chr (6): strain, experiment, t_muris_dose, lab_or_rw, cohort, source
## dbl (1): mouse_id
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

```


data cleaning and organization

Four original cell culture plates, each stimulated on a different day:

Original plate 1 stimulated July 22, 2025 -> experiment_day = 1

Original plate 2 stimulated July 23, 2025 -> experiment_day = 2

Original plate 3 stimulated July 29, 2025 -> experiment_day = 3

Original plate 4 stimulated July 30, 2025 -> experiment_day = 4

Each mouse appears on exactly one stimulation plate (one experiment_day).

For ELISA detection, original plates were combined into 2 detection plates:

Detection plate 1 = original plates 1 + 2 (July 22 + 23)

Detection plate 2 = original plates 3 + 4 (July 29 + 30)

The detection plate number is coded as plate in the cleaning chunks below.

experiment_day (1-4) must be added as a column in the plate map Excel files

so that it joins in automatically via left_join(plate1_map/plate2_map).

ifn data cleaning for plate 1

```
clean_ifn_1 <- elisa_ifn_1 %>%
  clean_names() %>%
  rename("final_ifn" = "final_if_ng") %>% # cleaner names
  left_join(plate1_map, by = "well_id") %>% # join elisa data and mouse number using plate number
  mutate(plate = 1) %>% # create a new column to signal that this is plate 1
  left_join(meta_data, by = "mouse_id") %>% # add the rest of the meta data
  mutate(
    across(c(concentration,
```

```

      final_ifn, mean,
      std_dev,
      cv_percent),
# apply the same transformation across these columns
  ~ case_when(
    # ~ is like a mini function. case when is checking for a condition and applying the first
    str_detect(., "^<") ~ 0,
    # . is the current value in the column and the rest of the function is asking if this value
    TRUE ~ as.numeric(str_remove(., "^>"))
  ) # the TRUE means that if none of the previous conditions matched, then do these. Same logic
) %>%
drop_na(mouse_id)

```

```

## Warning: There were 5 warnings in 'mutate()'.
## The first warning was:
## i In argument: 'across(...)'.
## Caused by warning:
## ! NAs introduced by coercion
## i Run 'dplyr::last_dplyr_warnings()' to see the 4 remaining warnings.

```

ifn data cleaning for plate 2

```

clean_ifn_2 <- elisa_ifn_2 %>% #same as before...
  clean_names() %>%
  rename("final_ifn" = "final_ifng") %>%
  left_join(plate2_map, by = "well_id") %>%
  mutate(plate = 2) %>%
  left_join(meta_data, by = "mouse_id") %>%
  mutate(
    across(c(concentration, final_ifn, mean, std_dev, cv_percent),
      ~ case_when(
        str_detect(., "^<") ~ 0,
        TRUE ~ as.numeric(str_remove(., "^>"))
      )
    )
  ) %>%
  drop_na(mouse_id)

```

```

## Warning: There were 5 warnings in 'mutate()'.
## The first warning was:
## i In argument: 'across(...)'.
## Caused by warning:
## ! NAs introduced by coercion
## i Run 'dplyr::last_dplyr_warnings()' to see the 4 remaining warnings.

```

```

total_ifn <- bind_rows(clean_ifn_1,
  clean_ifn_2)

write_csv(total_ifn, file.path(clean_data_path, "combined_ifn_elisa_data.csv"))

```

For now, I just kept all of the values that are over the detection limit without their “>” and all of the small values as zero. Discuss with David.

tnf data cleaning for plate 1

```
clean_tnf_1 <- elisa_tnf_1 %>% #same as before...
  clean_names() %>%
  rename("final_tnf" = "concentration") %>%
  left_join(plate1_map, by = "well_id") %>%
  mutate(plate = 1) %>%
  left_join(meta_data, by = "mouse_id") %>%
  mutate(
    across(c(final_tnf, mean, std_dev, cv_percent),
      ~ case_when(
        str_detect(., "<") ~ 0,
        TRUE ~ as.numeric(str_remove(., ">"))
      )
    ) %>%
  drop_na(mouse_id)
```

```
## Warning: There were 4 warnings in 'mutate()'.
## The first warning was:
## i In argument: 'across(...)'.
## Caused by warning:
## ! NAs introduced by coercion
## i Run 'dplyr::last_dplyr_warnings()' to see the 3 remaining warnings.
```

tnf data cleaning for plate 2

```
clean_tnf_2 <- elisa_tnf_2 %>% #same as before...
  clean_names() %>%
  rename("final_tnf" = "concentration") %>%
  left_join(plate2_map, by = "well_id") %>%
  mutate(plate = 2) %>%
  left_join(meta_data, by = "mouse_id") %>%
  mutate(
    across(c(final_tnf, mean, std_dev, cv_percent),
      ~ case_when(
        str_detect(., "<") ~ 0,
        TRUE ~ as.numeric(str_remove(., ">"))
      )
    ) %>%
  drop_na(mouse_id)
```

```
## Warning: There were 4 warnings in 'mutate()'.
## The first warning was:
## i In argument: 'across(...)'.
## Caused by warning:
## ! NAs introduced by coercion
## i Run 'dplyr::last_dplyr_warnings()' to see the 3 remaining warnings.
```

```
total_tnf <- bind_rows(clean_tnf_1,
                       clean_tnf_2)

write_csv(total_tnf, file.path(clean_data_path,
                              "combined_tnf_elisa_data.csv"))
```

Calculating the mean concentration value

```
ifn_input <- mean_calculator(total_ifn,
                             final_ifn,
                             cv_percent) %>%
  mutate(treatment = factor(treatment, levels = c("CD28", "LPS", "IgG")))

tnf_input <- mean_calculator(total_tnf,
                             final_tnf,
                             cv_percent) %>%
  mutate(treatment = factor(treatment, levels = c("CD28", "LPS", "IgG")))
```

I wonder if I should use the original data points instead of the mean data.

Filter the data for treatment specific plots

```
ifn_input_cd28 <- ifn_input %>% filter(treatment == "CD28")
ifn_input_lps <- ifn_input %>% filter(treatment == "LPS")
ifn_input_igg <- ifn_input %>% filter(treatment == "IgG")
tnf_input_cd28 <- tnf_input %>% filter(treatment == "CD28")
tnf_input_lps <- tnf_input %>% filter(treatment == "LPS")
tnf_input_igg <- tnf_input %>% filter(treatment == "IgG")
```

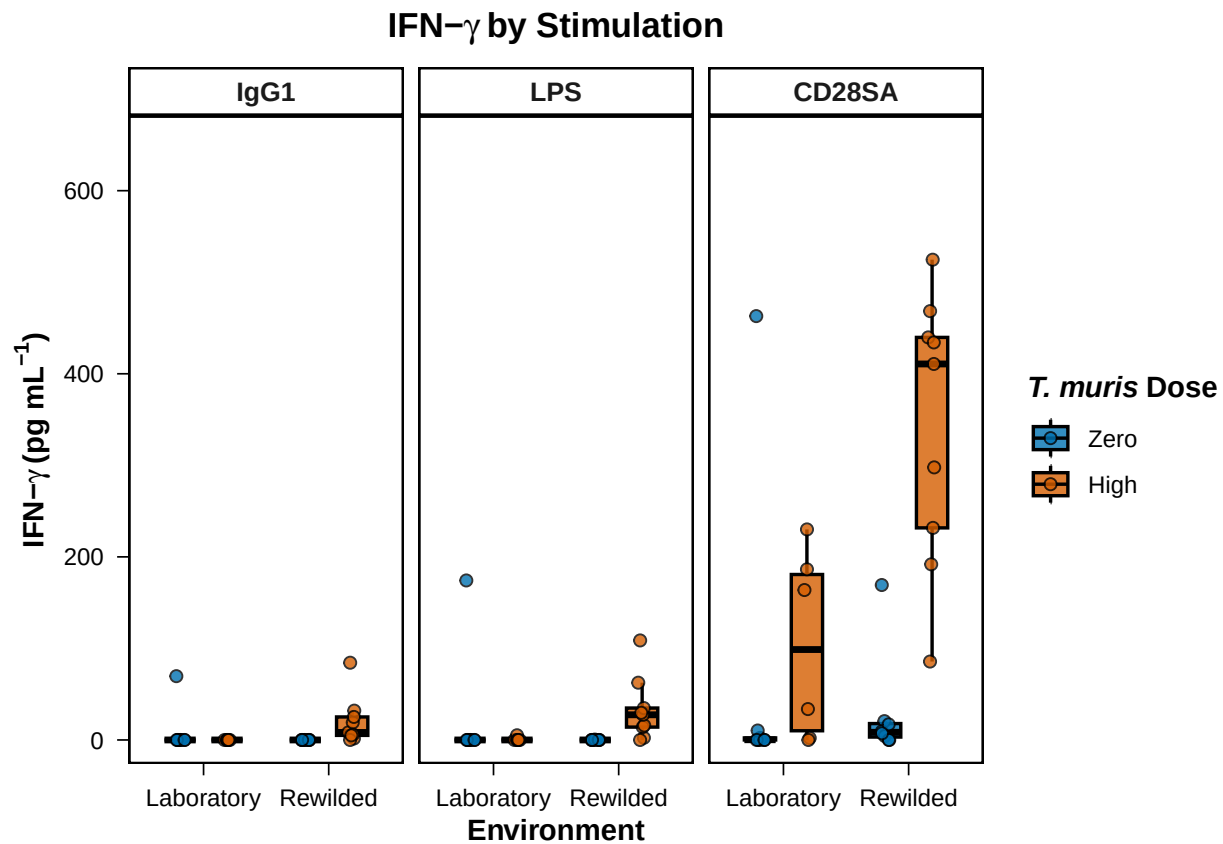
General ifn plot to get an overview

```
ifn_input <- ifn_input %>%
  mutate(t_muris_dose = factor(t_muris_dose,
                              levels = c("Zero", "High")))

ifn_plot <- elisa_summary_plotter(ifn_input,
                                 expression(bold("IFN-" * gamma ~ "(pg mL-1*)")),
                                 title = expression(bold("IFN-*gamma~"by Stimulation"))))

ifn_plot
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```



```
plot_saver("general_ifn_plot.png",
  ifn_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

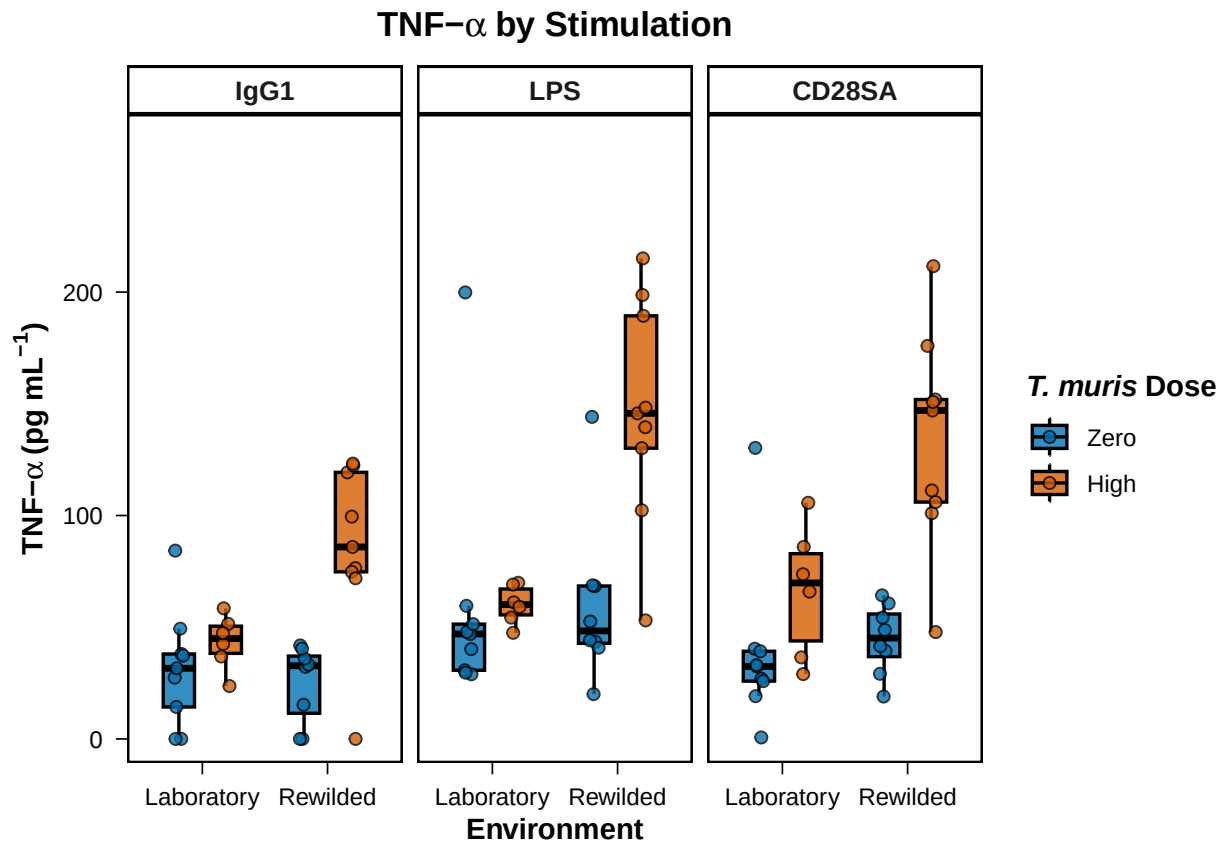
General tnf plots

```
tnf_input <- tnf_input %>%
  mutate(t_muris_dose = factor(t_muris_dose,
    levels = c("Zero", "High")))

tnf_plot <- elisa_summary_plotter(tnf_input,
  expression(bold("TNF-*alpha ~ "(pg mL-1*)")),
  title = expression(bold("TNF-*alpha~"by Stimulation))))

tnf_plot
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```



```
plot_saver("general_tnf_plot.png",
  tnf_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

IFN faceted by T. muris dose: treatment on x, environment as color

```
ifn_dose_facet_plot <- ggplot(ifn_input %>%
  mutate(treatment = factor(treatment,
    levels = c("IgG",
      "LPS",
      "CD28"))),
  aes(x = treatment,
    y = mean_concentration,
    fill = lab_or_rw)) +
  geom_boxplot(position = position_dodge(width = 0.75),
    alpha = 0.8,
    outlier.shape = NA,
    width = 0.5,
    color = "black",
```

```

        linewidth = 0.6) +
geom_jitter(position = position_jitterdodge(jitter.width = 0.15,
                                             dodge.width = 0.75),
            alpha = 0.8,
            size = 1.8,
            shape = 21,
            color = "black") +
facet_wrap(~ t_muris_dose,
           labeller = as_labeller(c("Zero" = "Zero Dose",
                                     "High" = "High Dose")))) +
scale_fill_manual(values = c("Lab" = color_mustard,
                             "RW" = color_green),
                 labels = c("Lab" = "Laboratory",
                             "RW" = "Rewilded"),
                 name = expression(bold("Environment")))) +
scale_color_manual(values = c("Lab" = color_mustard,
                              "RW" = color_green),
                  labels = c("Lab" = "Laboratory",
                              "RW" = "Rewilded"),
                  name = expression(bold("Environment")))) +
scale_x_discrete(name = expression(bold("Treatment")),
                 labels = c("CD28" = "CD28SA",
                             "LPS" = "LPS",
                             "IgG" = "IgG1")) +
scale_y_continuous(name = expression(bold("IFN- $\gamma$  ~
                                     (pg mL-1 * ")")),
                  expand = expansion(mult = c(0.05, 0.3))) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                       color = "black",
                                       linewidth = 0.8),
      strip.text = element_text(size = 10,
                                face = "bold"),
      panel.border = element_rect(color = "black",
                                  fill = NA,
                                  linewidth = 0.8),
      panel.grid = element_blank(),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      axis.text = element_text(color = "black"),
      axis.ticks = element_line(color = "black",
                                linewidth = 0.3),
      axis.ticks.length = unit(0.15, "cm"),
      plot.title = element_text(hjust = 0.5, face = "bold")) +
ggtitle(expression(bold("IFN- $\gamma$  ~ "by Treatment and Dose"))))

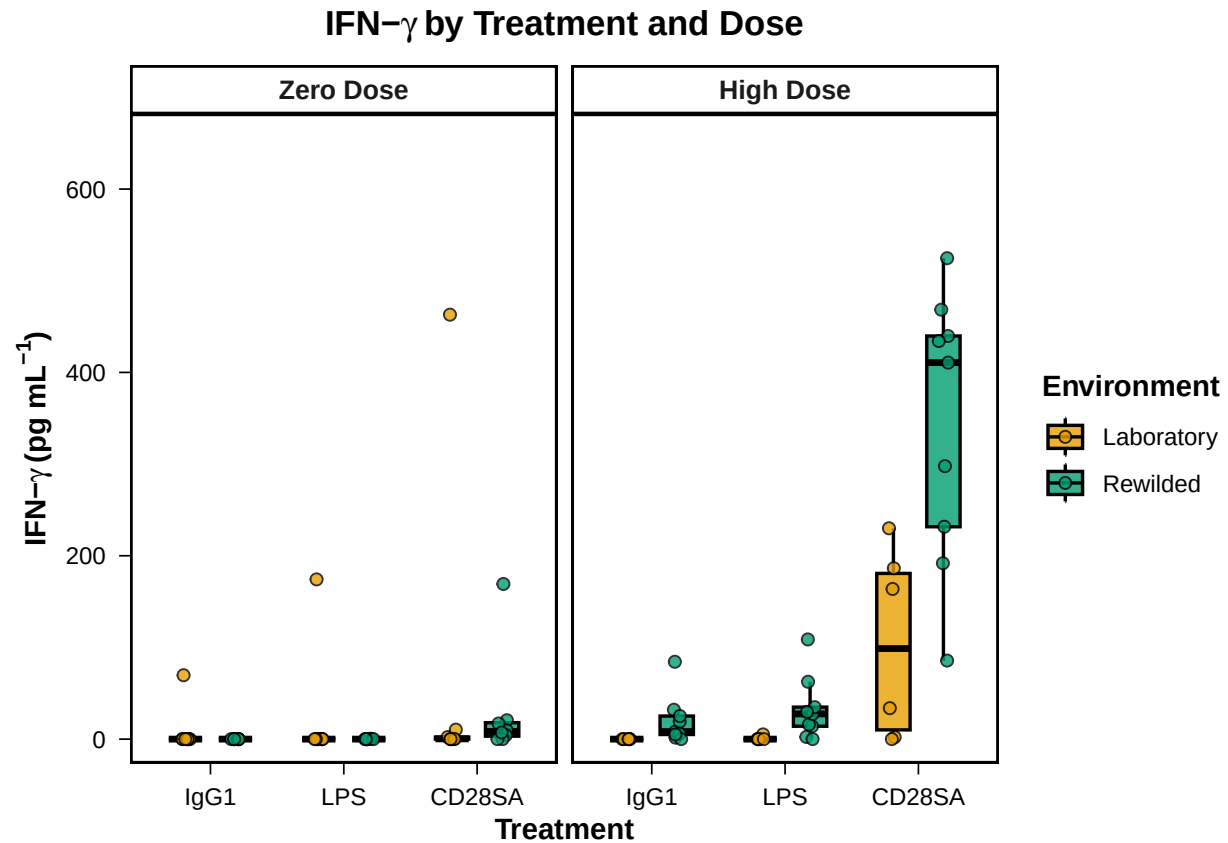
ifn_dose_facet_plot

```

```

## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.

```



```
plot_saver("ifn_dose_facet_plot.png",
  ifn_dose_facet_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

TNF faceted by T. muris dose: treatment on x, environment as color

```
tnf_dose_facet_plot <- ggplot(tnf_input %>%
  mutate(treatment = factor(treatment,
    levels = c("IgG",
      "LPS",
      "CD28"))),
  aes(x = treatment,
    y = mean_concentration,
    fill = lab_or_rw)) +
  geom_boxplot(position = position_dodge(width = 0.75),
    alpha = 0.8,
    outlier.shape = NA,
    width = 0.5,
    color = "black",
```



```

        linewidth = 0.6) +
geom_jitter(position = position_jitterdodge(jitter.width = 0.15,
                                             dodge.width = 0.75),

            alpha = 0.8,
            size = 1.8,
            shape = 21,
            color = "black") +
facet_wrap(~ t_muris_dose,
           labeller = as_labeller(c("Zero" = "Zero Dose",
                                     "High" = "High Dose")))) +
scale_fill_manual(values = c("Lab" = color_mustard,
                             "RW" = color_green),
                 labels = c("Lab" = "Laboratory",
                             "RW" = "Rewilded"),
                 name = expression(bold("Environment")))) +
scale_color_manual(values = c("Lab" = color_mustard,
                              "RW" = color_green),
                  labels = c("Lab" = "Laboratory",
                              "RW" = "Rewilded"),
                  name = expression(bold("Environment")))) +
scale_x_discrete(name = expression(bold("Treatment")),
                 labels = c("CD28" = "CD28SA",
                             "LPS" = "LPS",
                             "IgG" = "IgG1")) +
scale_y_continuous(name = expression(bold("TNF-" * alpha ~
                                         "(pg mL"^-1" * ")")),
                  expand = expansion(mult = c(0.05, 0.3))) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                       color = "black",
                                       linewidth = 0.8),

      strip.text = element_text(size = 10,
                                face = "bold"),
      panel.border = element_rect(color = "black",
                                  fill = NA,
                                  linewidth = 0.8),

      panel.grid = element_blank(),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      axis.text = element_text(color = "black"),
      axis.ticks = element_line(color = "black",
                                linewidth = 0.3),
      axis.ticks.length = unit(0.15, "cm"),
      plot.title = element_text(hjust = 0.5, face = "bold")) +
ggtitle(expression(bold("TNF-" * alpha ~ "by Treatment and Dose"))))

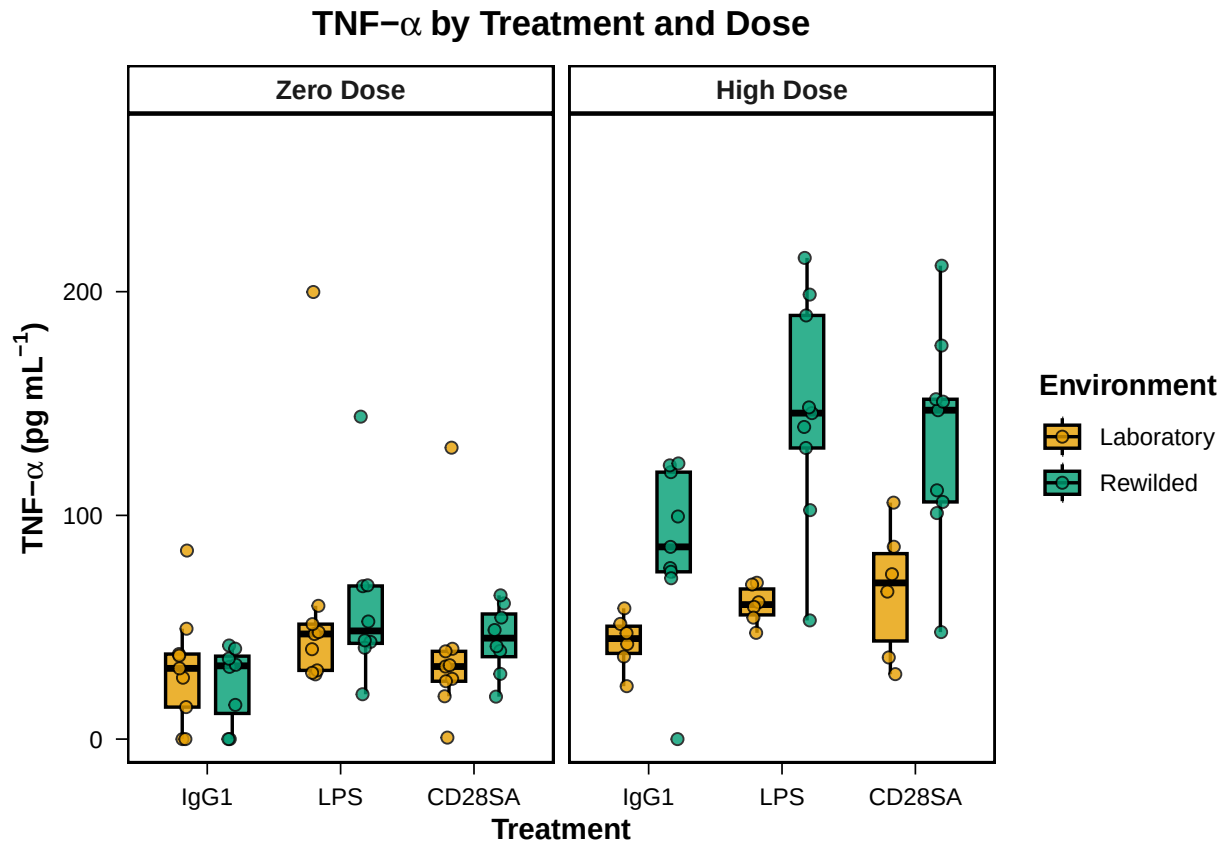
tnf_dose_facet_plot

```

```

## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.

```



```
plot_saver("tnf_dose_facet_plot.png",
  tnf_dose_facet_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

IFN faceted by environment: treatment on x, dose as color (#3)

```
ifn_env_facet_plot <- ggplot(ifn_input %>%
  mutate(treatment = factor(treatment,
    levels = c("IgG",
      "LPS",
      "CD28")),
  lab_or_rw = factor(lab_or_rw,
    levels = c("Lab", "RW")),
  t_muris_dose = factor(t_muris_dose,
    levels = c("Zero", "High"))),
  aes(x = treatment,
    y = mean_concentration,
    fill = t_muris_dose)) +
  geom_boxplot(position = position_dodge(width = 0.75),
    alpha = 0.8,
```

```

    outlier.shape = NA,
    width = 0.5,
    color = "black",
    linewidth = 0.6) +
geom_jitter(position = position_jitterdodge(jitter.width = 0.15,
                                           dodge.width = 0.75),

    alpha = 0.8,
    size = 1.8,
    shape = 21,
    color = "black") +
facet_wrap(~ lab_or_rw,
    labeller = as_labeller(c("Lab" = "Laboratory",
                             "RW" = "Rewilded")))) +
scale_fill_manual(values = c("Zero" = color_dark_blue,
                             "High" = color_orange),
    name = expression(bolditalic("T. muris") ~ bold("Dose")))) +
scale_color_manual(values = c("Zero" = color_dark_blue,
                             "High" = color_orange),
    name = expression(bolditalic("T. muris") ~ bold("Dose")))) +
scale_x_discrete(name = expression(bold("Treatment")),
    labels = c("CD28" = "CD28SA",
               "LPS" = "LPS",
               "IgG" = "IgG1")) +
scale_y_continuous(name = expression(bold("IFN-" * gamma ~
                                         "(pg mL"^-1" * ")")),
    expand = expansion(mult = c(0.05, 0.3))) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                       color = "black",
                                       linewidth = 0.8),
    strip.text = element_text(size = 10,
                              face = "bold"),
    panel.border = element_rect(color = "black",
                              fill = NA,
                              linewidth = 0.8),
    panel.grid = element_blank(),
    axis.title.x = element_text(face = "bold"),
    axis.title.y = element_text(face = "bold"),
    axis.text = element_text(color = "black"),
    axis.ticks = element_line(color = "black",
                              linewidth = 0.3),
    axis.ticks.length = unit(0.15, "cm"),
    plot.title = element_text(hjust = 0.5, face = "bold")) +
ggtitle(expression(bold("IFN-" * gamma ~ "by Treatment and Environment"))))

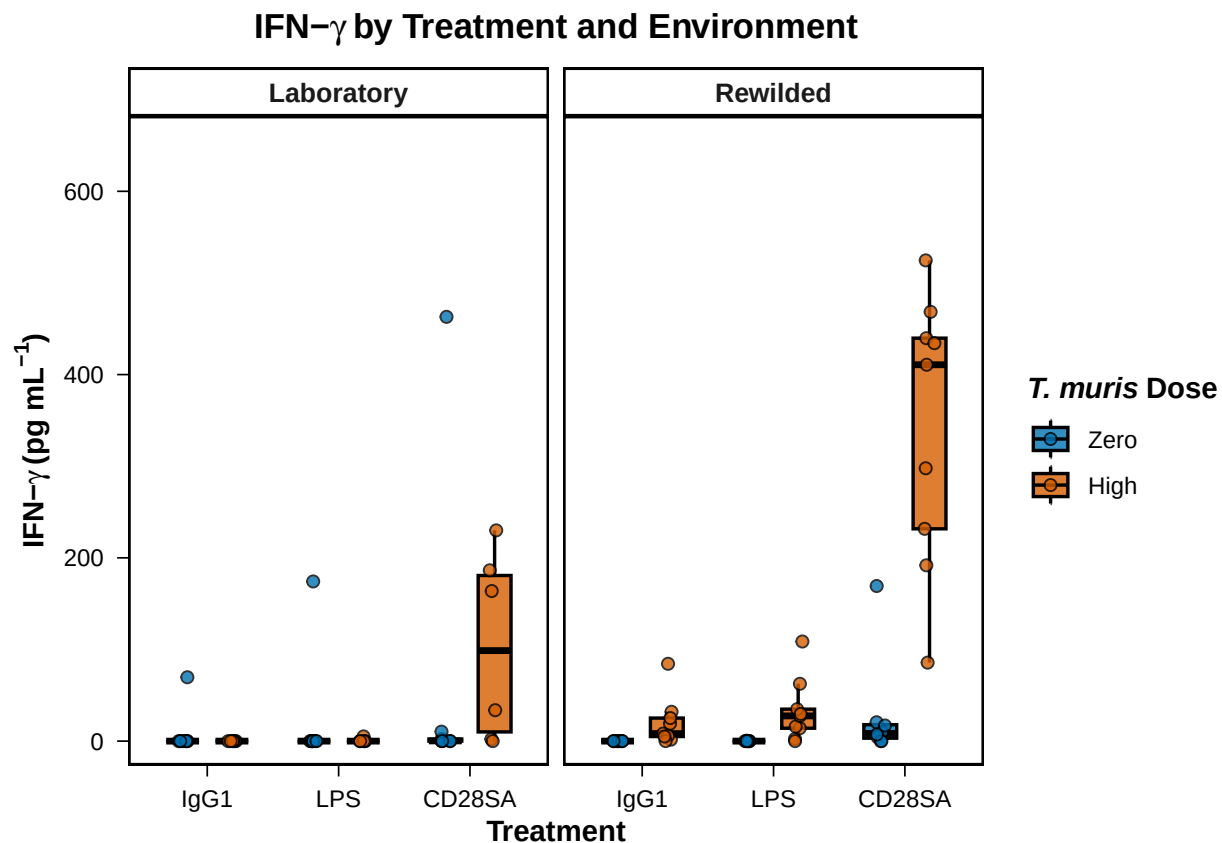
ifn_env_facet_plot

```

```

## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.

```



```
plot_saver("ifn_env_facet_plot.png",
  ifn_env_facet_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

TNF faceted by environment: treatment on x, dose as color (#3)

```
tnf_env_facet_plot <- ggplot(tnf_input %>%
  mutate(treatment = factor(treatment,
    levels = c("IgG",
               "LPS",
               "CD28")),
  lab_or_rw = factor(lab_or_rw,
    levels = c("Lab", "RW")),
  t_muris_dose = factor(t_muris_dose,
    levels = c("Zero", "High"))),
  aes(x = treatment,
    y = mean_concentration,
    fill = t_muris_dose)) +
  geom_boxplot(position = position_dodge(width = 0.75),
    alpha = 0.8,
```

```

    outlier.shape = NA,
    width = 0.5,
    color = "black",
    linewidth = 0.6) +
geom_jitter(position = position_jitterdodge(jitter.width = 0.15,
                                           dodge.width = 0.75),

    alpha = 0.8,
    size = 1.8,
    shape = 21,
    color = "black") +
facet_wrap(~ lab_or_rw,
    labeller = as_labeller(c("Lab" = "Laboratory",
                             "RW" = "Rewilded")))) +
scale_fill_manual(values = c("Zero" = color_dark_blue,
                             "High" = color_orange),
    name = expression(bolditalic("T. muris") ~ bold("Dose")))) +
scale_color_manual(values = c("Zero" = color_dark_blue,
                             "High" = color_orange),
    name = expression(bolditalic("T. muris") ~ bold("Dose")))) +
scale_x_discrete(name = expression(bold("Treatment")),
    labels = c("CD28" = "CD28SA",
               "LPS" = "LPS",
               "IgG" = "IgG1")) +
scale_y_continuous(name = expression(bold("TNF-" * alpha ~
                                       "(pg mL"^-1" * ")")),
    expand = expansion(mult = c(0.05, 0.3))) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                       color = "black",
                                       linewidth = 0.8),
    strip.text = element_text(size = 10,
                              face = "bold"),
    panel.border = element_rect(color = "black",
                                fill = NA,
                                linewidth = 0.8),
    panel.grid = element_blank(),
    axis.title.x = element_text(face = "bold"),
    axis.title.y = element_text(face = "bold"),
    axis.text = element_text(color = "black"),
    axis.ticks = element_line(color = "black",
                              linewidth = 0.3),
    axis.ticks.length = unit(0.15, "cm"),
    plot.title = element_text(hjust = 0.5, face = "bold")) +
ggtitle(expression(bold("TNF-" * alpha ~ "by Treatment and Environment"))))

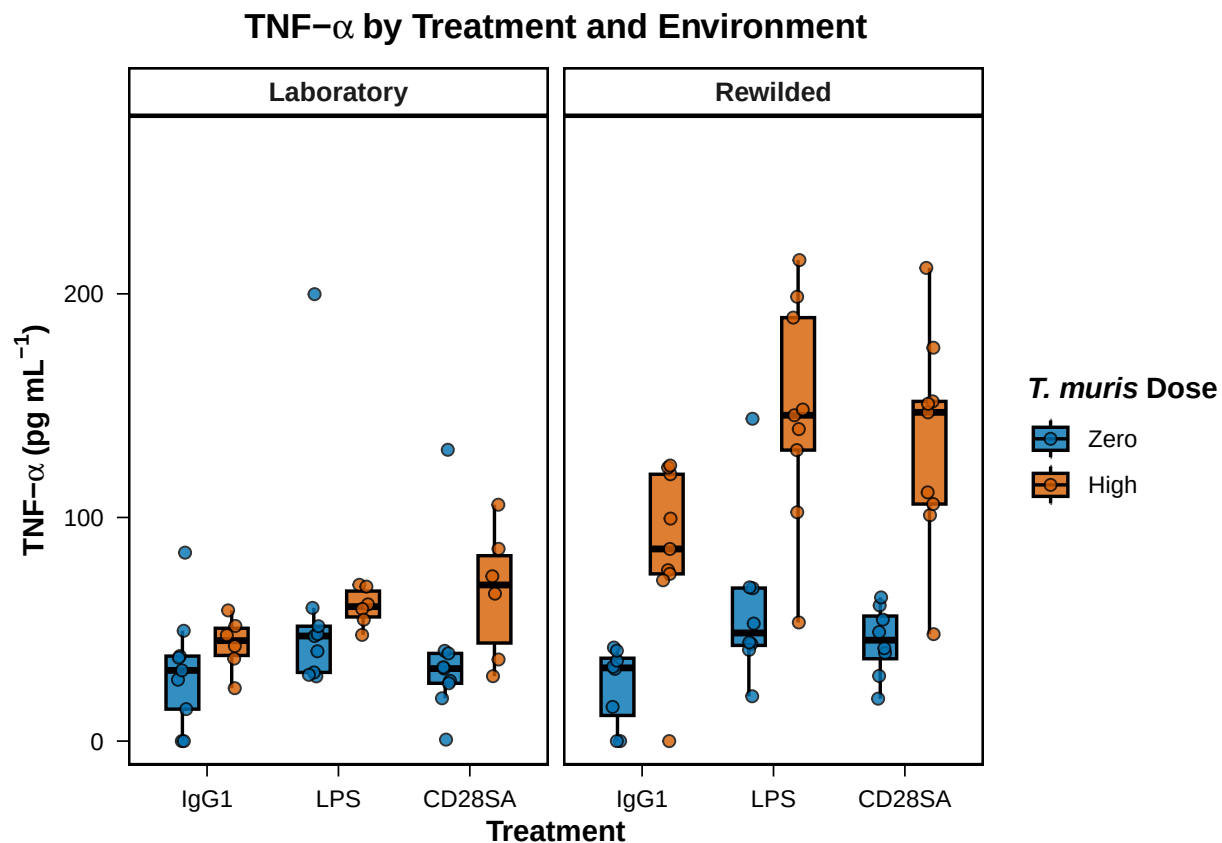
tnf_env_facet_plot

```

```

## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.

```



```
plot_saver("tnf_env_facet_plot.png",
  tnf_env_facet_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

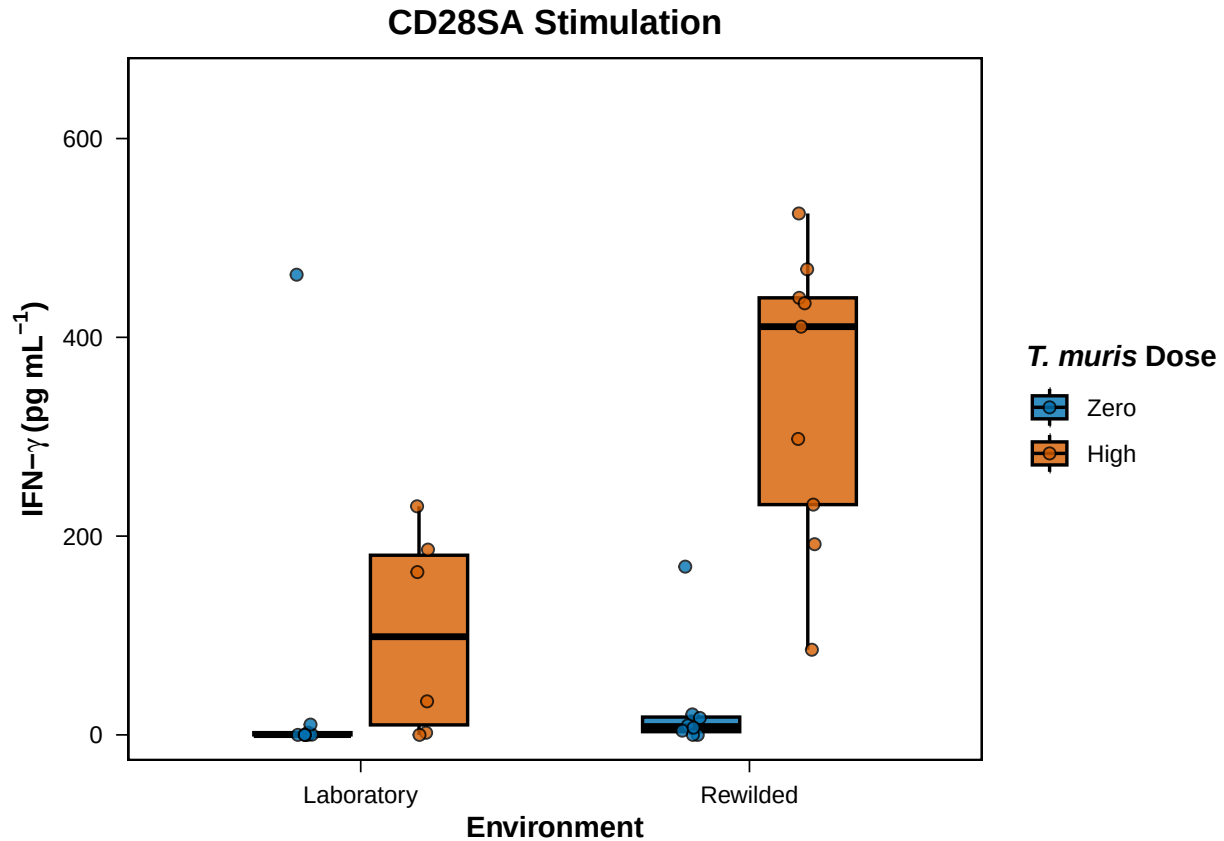
Treatment specific plots for ifn: CD28

```
ifn_input_cd28 <- ifn_input_cd28 %>%
  mutate(t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High")))

ifn_cd28_plot <- elisa_treatment_plotter(ifn_input_cd28,
  expression(bold("IFN-" * gamma ~ "(pg mL-1*)")),
  title = expression(bold("CD28SA Stimulation")))

ifn_cd28_plot
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```



```
plot_saver("ifn_cd28_treatment_plot.png",
  ifn_cd28_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

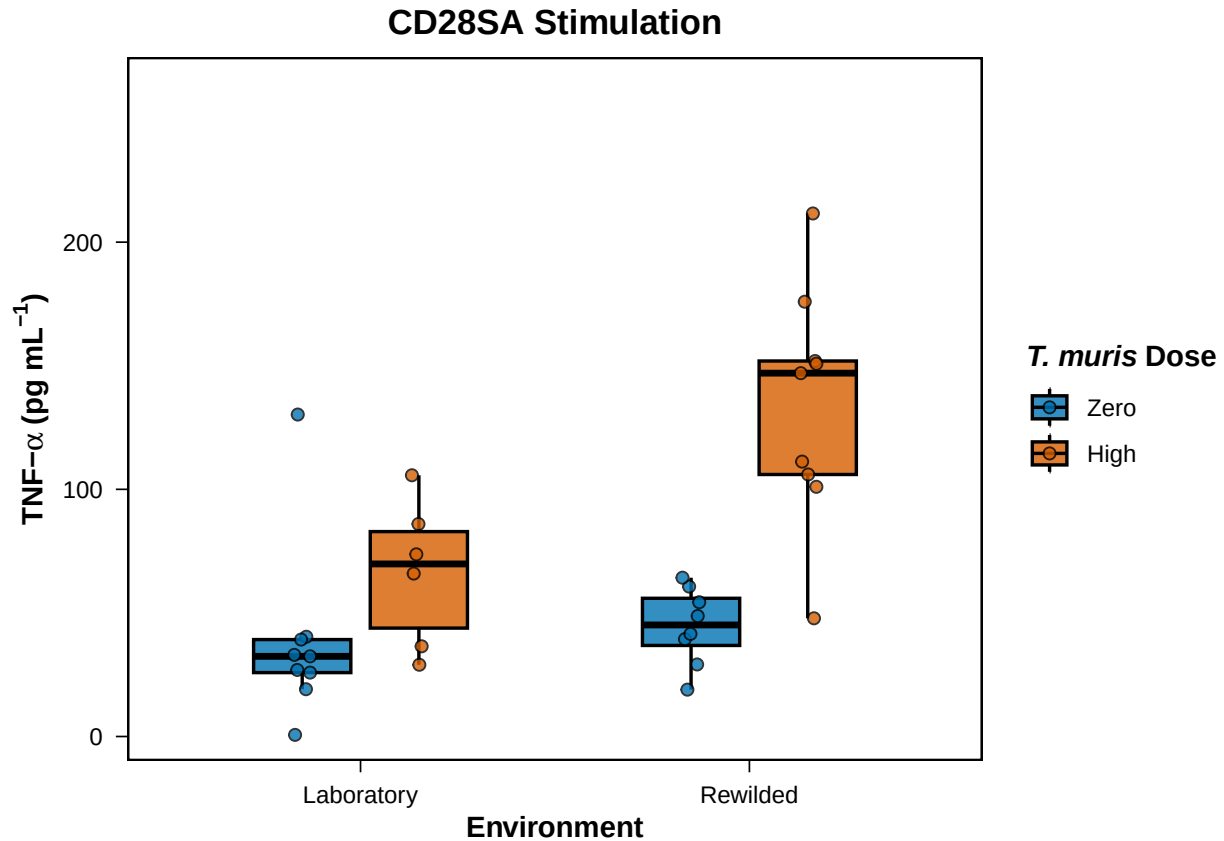
Treatment specific plots for tnfr: CD28

```
tnf_input_cd28 <- tnf_input_cd28 %>%
  mutate(t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High")))

tnf_cd28_plot <- elisa_treatment_plotter(tnf_input_cd28,
  expression(bold("TNF-*alpha ~ "(pg mL-1)*")),
  title = expression(bold("CD28SA Stimulation")))

tnf_cd28_plot
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```



```
plot_saver("tnf_cd28_treatment_plot.png",
  tnf_cd28_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

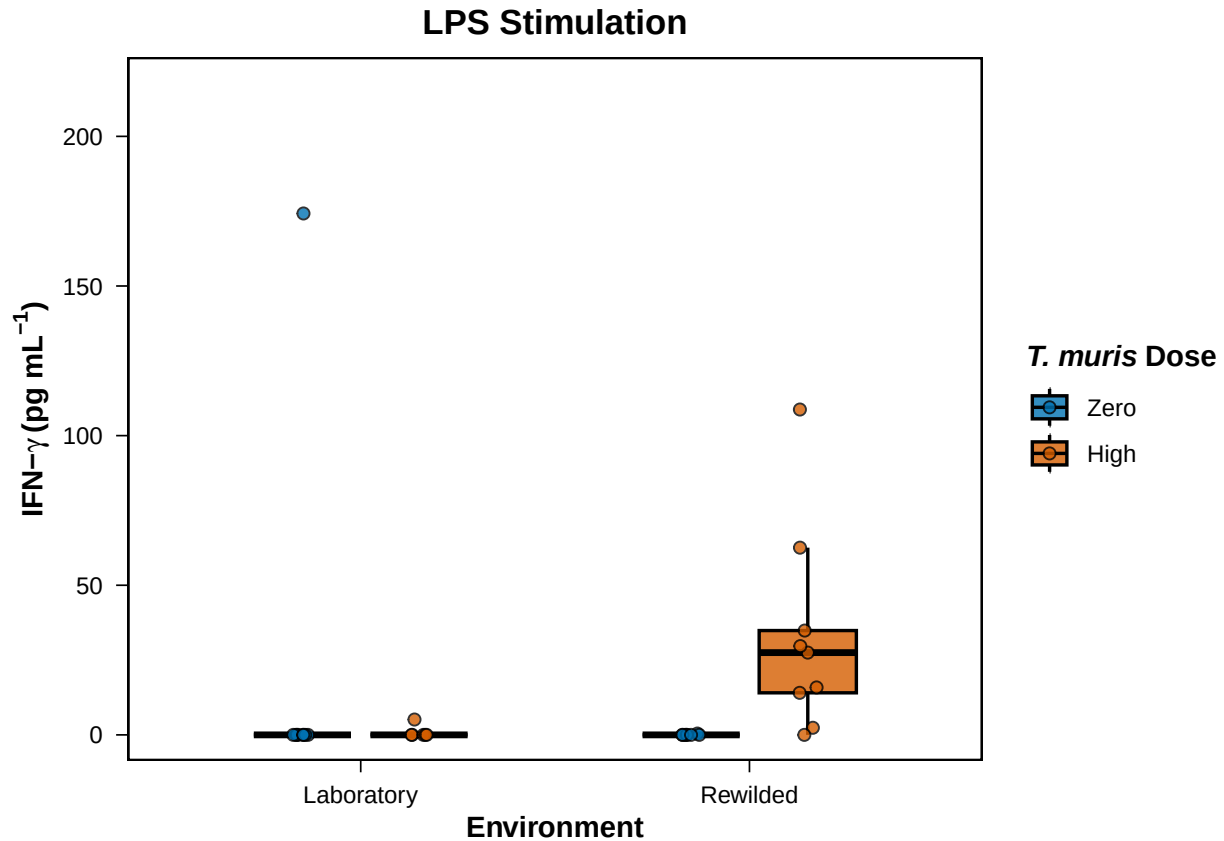
Treatment specific plots for ifn: LPS

```
ifn_input_lps <- ifn_input_lps %>%
  mutate(t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High")))

ifn_lps_plot <- elisa_treatment_plotter(ifn_input_lps,
  expression(bold("IFN-" * gamma ~ "(pg mL-1*)")),
  title = expression(bold("LPS Stimulation")))

ifn_lps_plot
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

```
plot_saver("ifn_lps_treatment_plot.png",
  ifn_lps_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

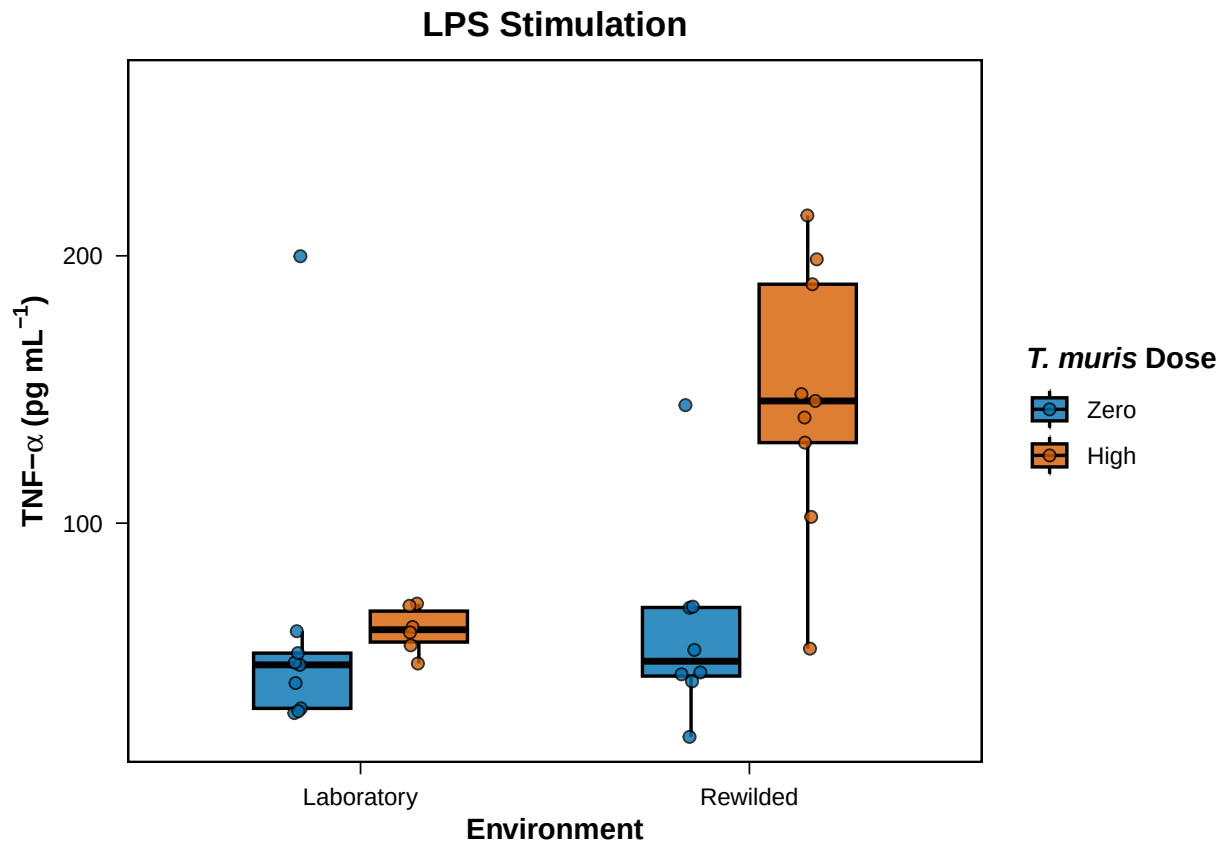
Treatment specific plots for tnfr: LPS

```
tnf_input_lps <- tnf_input_lps %>%
  mutate(t_muris_dose = factor(t_muris_dose,
    levels = c("Zero", "High")))

tnf_lps_plot <- elisa_treatment_plotter(tnf_input_lps,
  expression(bold("TNF-*alpha ~ "(pg mL)^-1"*")),
  title = expression(bold("LPS Stimulation")))

tnf_lps_plot
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```



```
plot_saver("tnf_lps_treatment_plot.png",
  tnf_lps_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

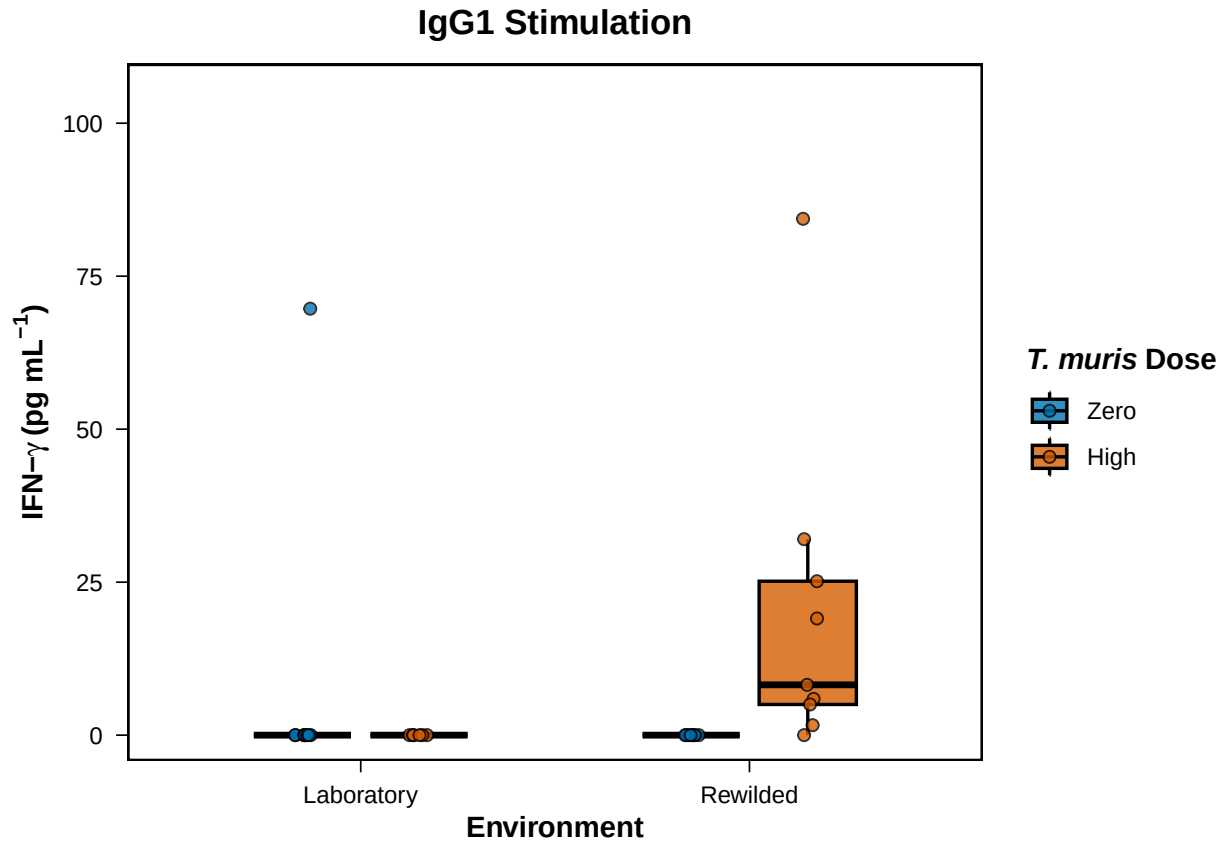
Treatment specific plots for ifn: IgG

```
ifn_input_igg <- ifn_input_igg %>%
  mutate(t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High")))

ifn_igg_plot <- elisa_treatment_plotter(ifn_input_igg,
  expression(bold("IFN-" * gamma ~ "(pg mL-1*)")),
  title = expression(bold("IgG1 Stimulation")))

ifn_igg_plot
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```



```
plot_saver("ifn_igg_treatment_plot.png",
  ifn_igg_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

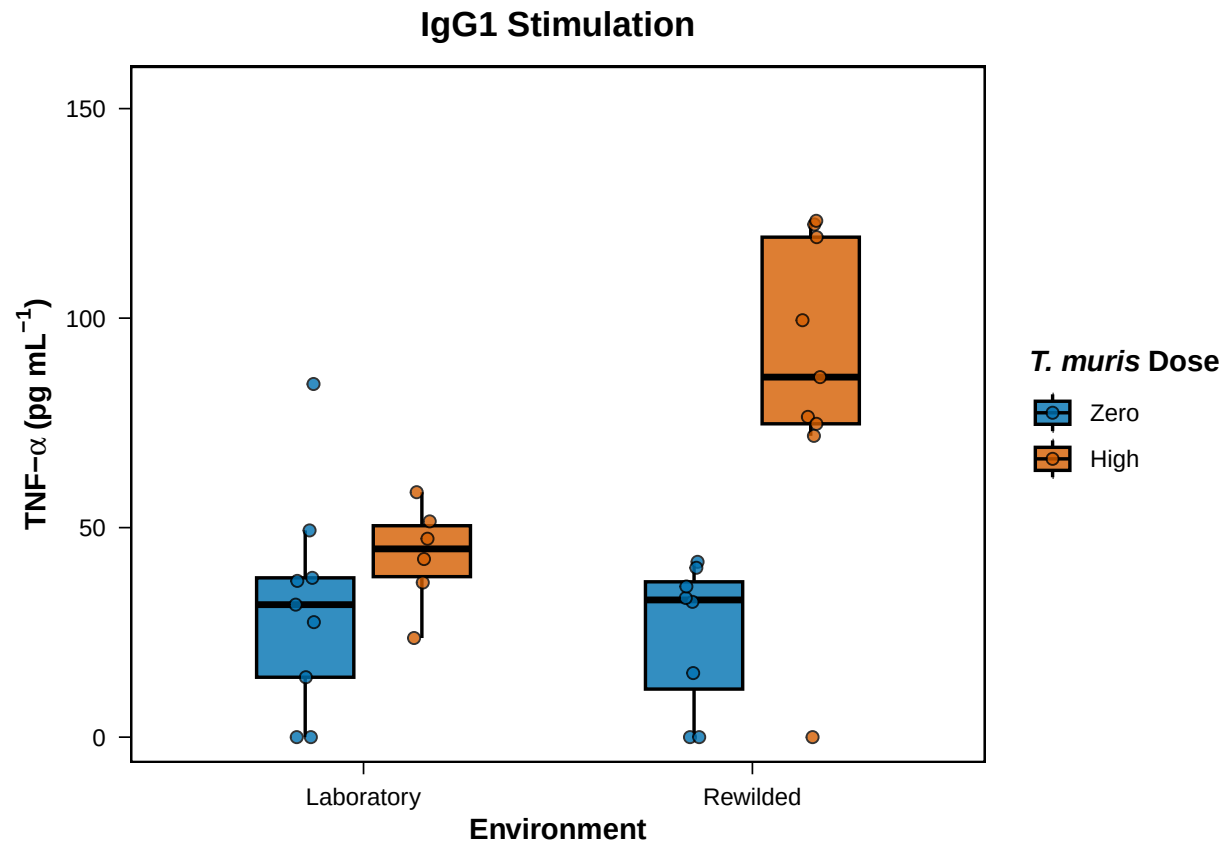
Treatment specific plots for tnfr: IgG

```
tnf_input_igg <- tnf_input_igg %>%
  mutate(t_muris_dose = factor(t_muris_dose,
    levels = c("Zero", "High")))

tnf_igg_plot <- elisa_treatment_plotter(tnf_input_igg,
  expression(bold("TNF-*alpha ~ "(pg mL)^-1"*")),
  title = expression(bold("IgG1 Stimulation")))

tnf_igg_plot
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```



```
plot_saver("tnf_igg_treatment_plot.png",  
           tnf_igg_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the  
## data's colour values.
```

Setting reference levels for ifn

Mixed effect model (gamma) for ifn and diagnostics

Note to self: should I just assume that all zeros are at the detection level?

Maybe we can just take half of the detection limit (standard practice)

Detection level for IFN is 1.797 pg/mL so we use this to handle zero values.

```
ifn_input <- ifn_input %>%
  mutate(
    t_muris_dose = relevel(factor(t_muris_dose), ref = "Zero"),
    treatment = relevel(factor(treatment), ref = "IgG"),
    lab_or_rw = relevel(factor(lab_or_rw), ref = "Lab")
  )

ifn_glmer_input <- ifn_input %>%
  mutate(mean_concentration = ifelse(mean_concentration == 0,
                                     1.797 / 2,
                                     mean_concentration))

ifn_glmer <- glmer(mean_concentration ~ lab_or_rw *
                  t_muris_dose *
                  treatment +
                  (1 | mouse_id) +
                  (1 | experiment_day),
                  family = Gamma(link = "log"),
                  control = glmerControl(optimizer = "bobyqa",
                                          optCtrl = list(maxfun = 2e5)),
                  data = ifn_glmer_input)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(ifn_glmer)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##   Family: Gamma ( log )
## Formula: mean_concentration ~ lab_or_rw * t_muris_dose * treatment + (1 |
##   mouse_id) + (1 | experiment_day)
##   Data: ifn_glmer_input
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
```

```

##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    605.0    643.5   -287.5    575.0      81
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.3141 -0.6229 -0.0637  0.3756  3.1796
##
## Random effects:
##   Groups             Name             Variance  Std.Dev.
##  mouse_id      (Intercept)  8.113e-01  9.007e-01
##  experiment_day (Intercept)  9.667e-10  3.109e-05
##  Residual                        5.550e-01  7.450e-01
## Number of obs: 96, groups:  mouse_id, 32; experiment_day, 4
##
## Fixed effects:
##
##                                     Estimate Std. Error t value Pr(>|z|)
## (Intercept)                        0.4149     0.3896   1.065 0.286938
## lab_or_rwRW                       -0.4207     0.5680  -0.741 0.458877
## t_muris_doseHigh                   -0.4167     0.6161  -0.676 0.498744
## treatmentCD28                      0.7130     0.3512   2.030 0.042328
## treatmentLPS                      0.1379     0.3512   0.393 0.694557
## lab_or_rwRW:t_muris_doseHigh       2.8889     0.8379   3.448 0.000566
## lab_or_rwRW:treatmentCD28          1.9120     0.5119   3.735 0.000188
## lab_or_rwRW:treatmentLPS          -0.2341     0.5119  -0.457 0.647436
## t_muris_doseHigh:treatmentCD28     3.5506     0.5553   6.394 1.61e-10
## t_muris_doseHigh:treatmentLPS      0.1336     0.5553   0.241 0.809866
## lab_or_rwRW:t_muris_doseHigh:treatmentCD28 -2.8198  0.7553  -3.734 0.000189
## lab_or_rwRW:t_muris_doseHigh:treatmentLPS  0.5800  0.7553   0.768 0.442521
##
## (Intercept)
## lab_or_rwRW
## t_muris_doseHigh
## treatmentCD28 *
## treatmentLPS
## lab_or_rwRW:t_muris_doseHigh ***
## lab_or_rwRW:treatmentCD28 ***
## lab_or_rwRW:treatmentLPS
## t_muris_doseHigh:treatmentCD28 ***
## t_muris_doseHigh:treatmentLPS
## lab_or_rwRW:t_muris_doseHigh:treatmentCD28 ***
## lab_or_rwRW:t_muris_doseHigh:treatmentLPS
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) lb__RW t_mr_H trCD28 trtLPS lb__RW:__H l__RW:C l__RW:L
## lab_or_rwRW -0.686
## t_mrs_dsHgh -0.632  0.434
## tretmntCD28 -0.451  0.309  0.285
## treatmntLPS -0.451  0.309  0.285  0.500
## lb_r_RW:__H  0.465 -0.678 -0.735 -0.210 -0.210
## lb__RW:CD28  0.309 -0.451 -0.196 -0.686 -0.343  0.305
## lb_r_RW:LPS  0.309 -0.451 -0.196 -0.343 -0.686  0.305      0.500

```

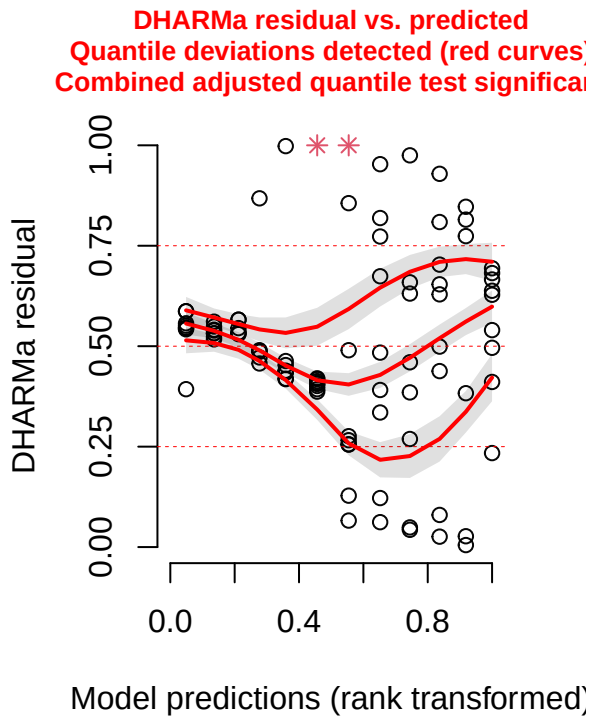
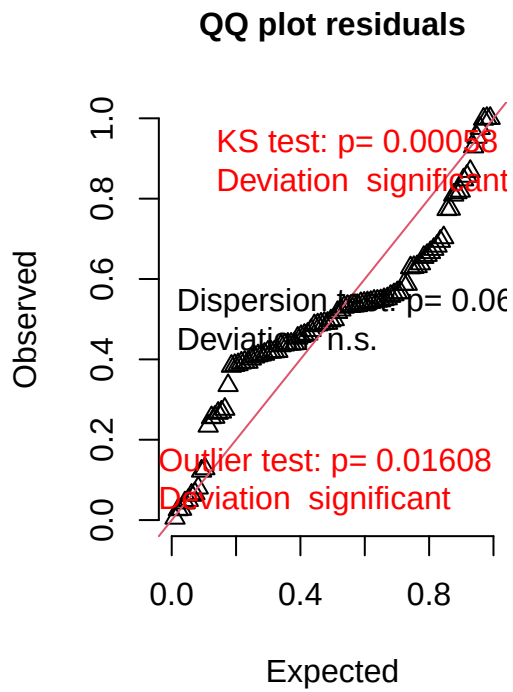
```
## t_mr_H:CD28  0.285 -0.196 -0.451 -0.632 -0.316  0.331      0.434  0.217
## t_mrs_H:LPS  0.285 -0.196 -0.451 -0.316 -0.632  0.331      0.217  0.434
## l__RW:__H:C -0.210  0.305  0.331  0.465  0.232 -0.451     -0.678 -0.339
## l__RW:__H:L -0.210  0.305  0.331  0.232  0.465 -0.451     -0.339 -0.678
##           t__H:C t__H:L l__RW:__H:C
## lab_or_rwRW
## t_mrs_dsHgh
## tretmntCD28
## treatmntLPS
## lb_r_RW:__H
## lb__RW:CD28
## lb_r_RW:LPS
## t_mr_H:CD28
## t_mrs_H:LPS  0.500
## l__RW:__H:C -0.735 -0.368
## l__RW:__H:L -0.368 -0.735  0.500
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

Diagnostics for ifn

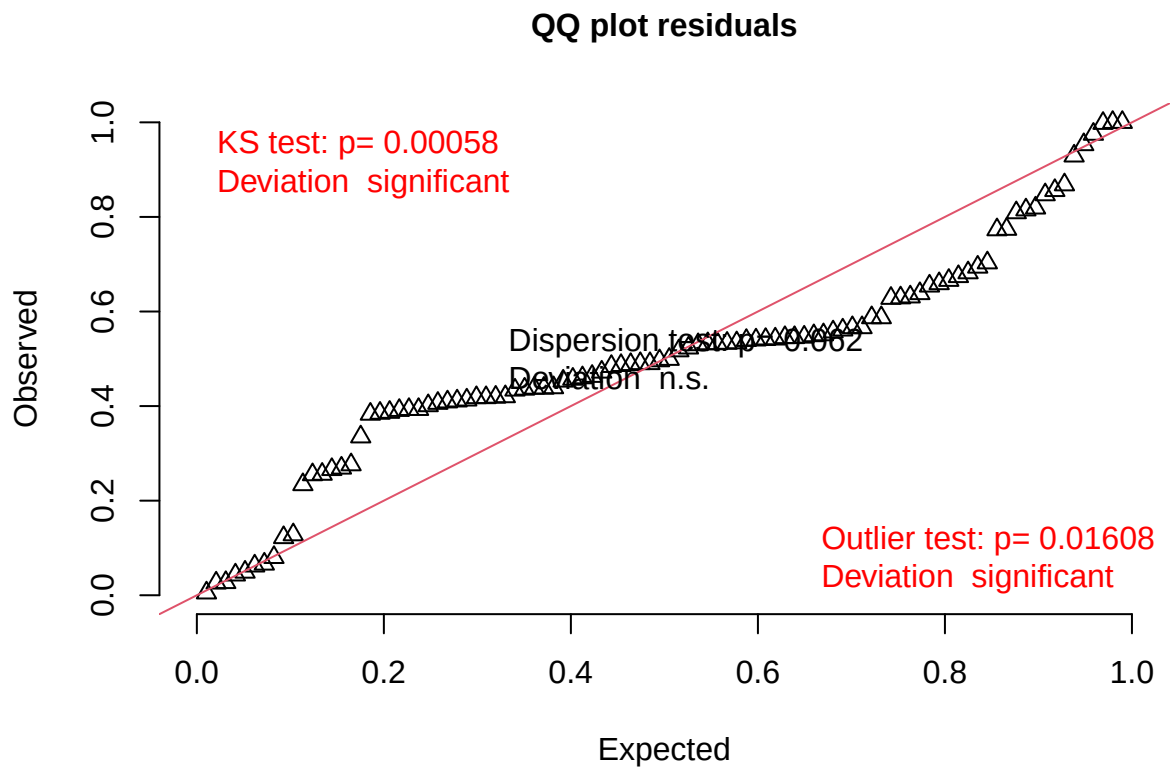
```
ifn_sim_res <- simulateResiduals(fittedModel = ifn_glmer, n = 1000)

plot(ifn_sim_res)
```

DHARMa residual



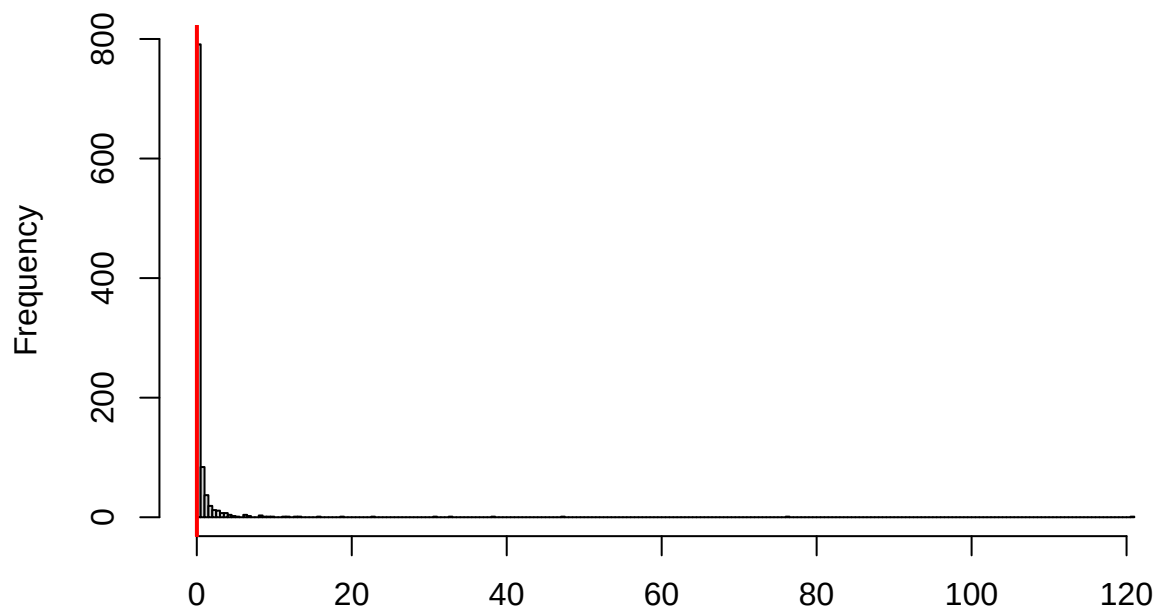
```
testUniformity(ifn_sim_res)
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.20592, p-value = 0.0005826
## alternative hypothesis: two-sided
```

```
testDispersion(ifn_sim_res)
```

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated

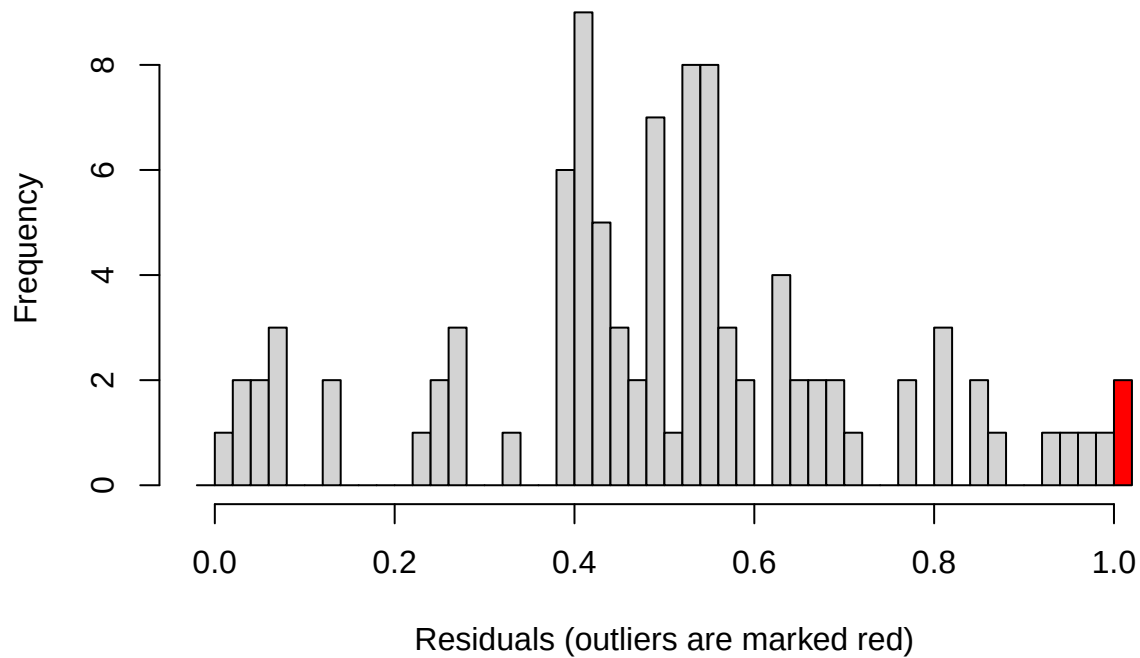


Simulated values, red line = fitted model. p-value (two.sided) = 0.062

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 0.015929, p-value = 0.062
## alternative hypothesis: two.sided
```

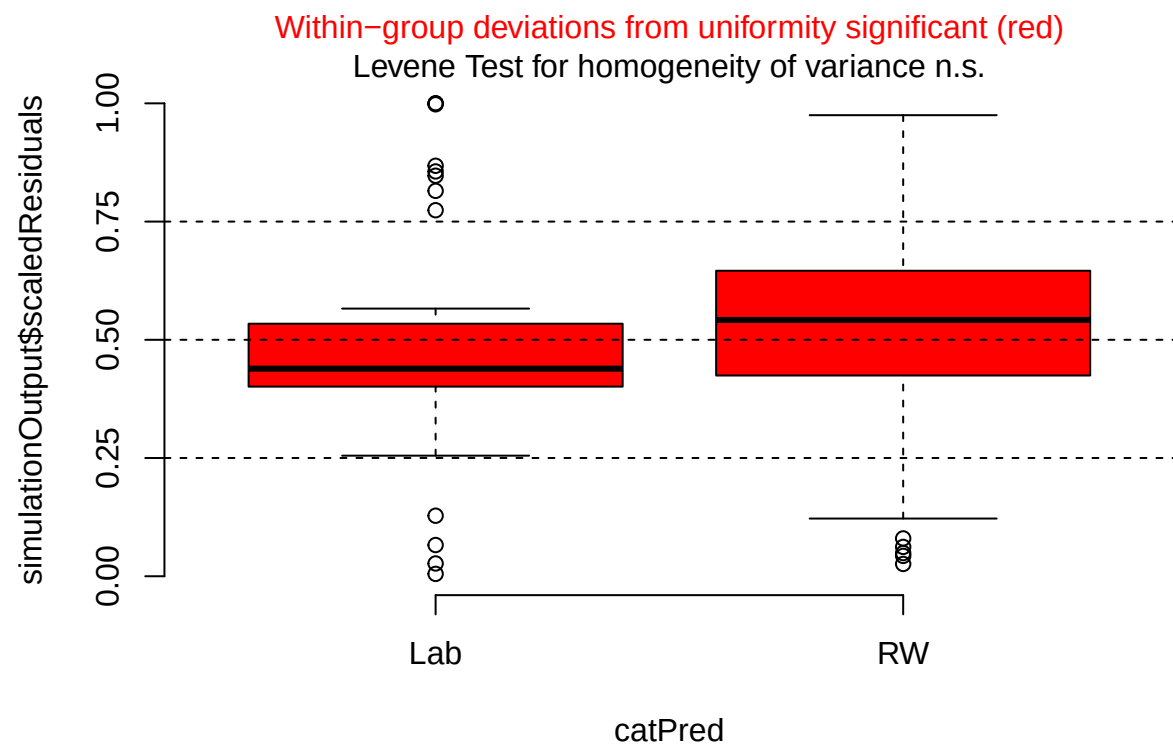
```
testOutliers(ifn_sim_res)
```

Outlier test significant

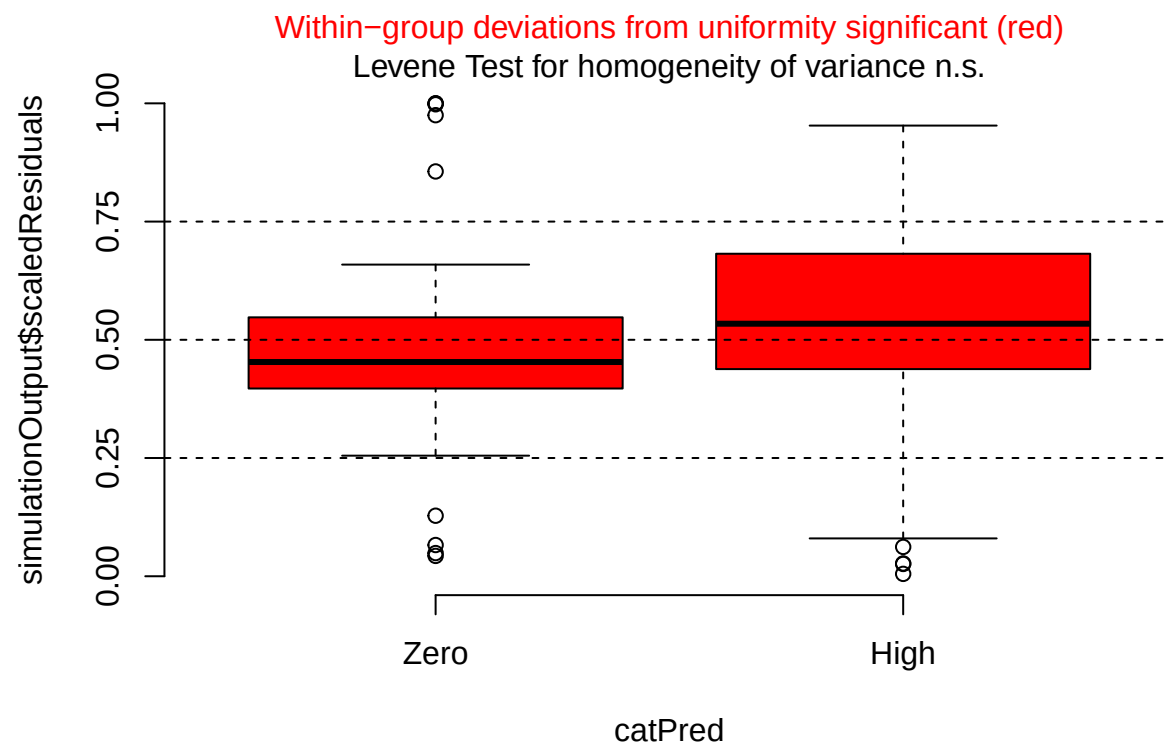


```
##
## DHARMa outlier test based on exact binomial test with approximate
## expectations
##
## data: ifn_sim_res
## outliers at both margin(s) = 2, observations = 96, p-value = 0.01608
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
## 0.002533047 0.073236831
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
## 0.02083333
```

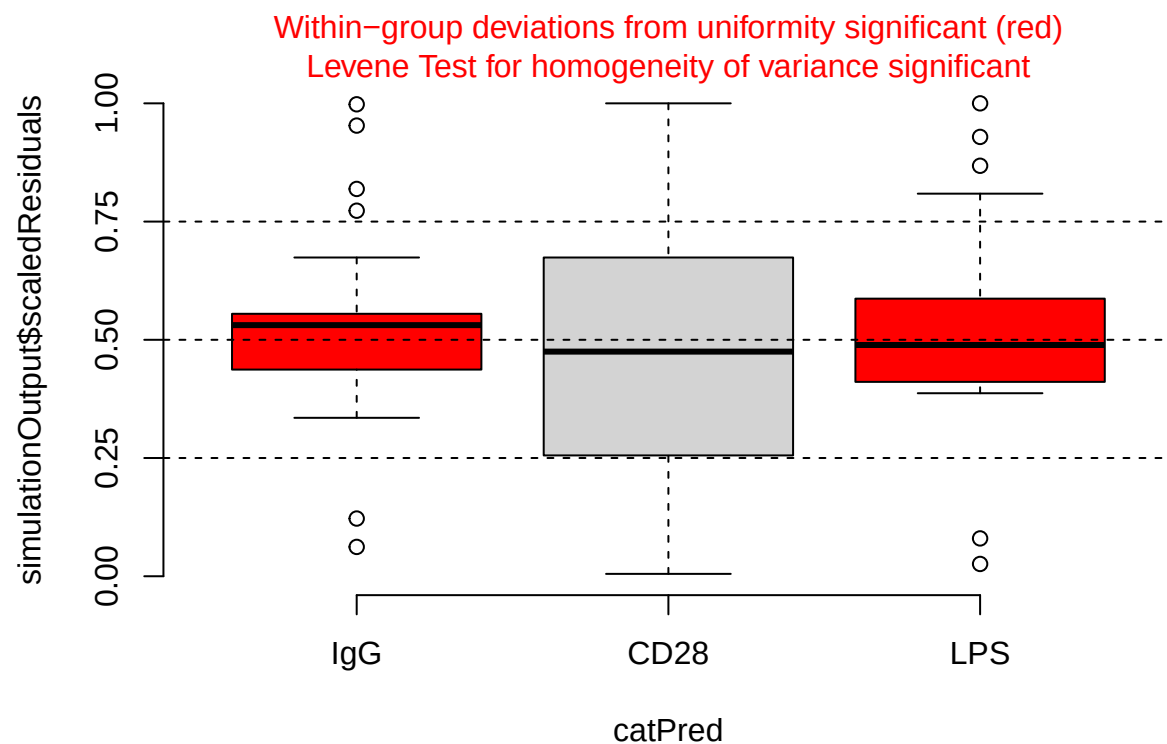
```
plotResiduals(ifn_sim_res, form = ifn_glmer_input$lab_or_rw)
```



```
plotResiduals(ifn_sim_res, form = ifn_glmer_input$t_muris_dose)
```



```
plotResiduals(ifn_sim_res, form = ifn_glmer_input$treatment)
```



```
ranef(ifn_glmer)
```

```
## $mouse_id
##      (Intercept)
## 1357  0.10187056
## 1359 -0.57711547
## 1360 -0.61121711
## 1361  0.21360103
## 1363  0.39674580
## 1364 -0.11902275
## 1366 -0.49788534
## 1367  1.29764204
## 1368  0.19653110
## 1370  0.81744506
## 1371 -0.45898926
## 1373  0.29982508
## 1374 -0.39402752
## 1376 -0.69454786
## 1378 -0.16479623
## 1383 -0.74002644
## 1385 -0.34226785
## 1388  0.32439506
## 1389  0.27831237
## 1391 -1.62653123
## 1392  0.03703968
```

```
## 1394 -0.30323294
## 1395 -0.34226785
## 1396  0.42234531
## 1398 -0.50652846
## 1401 -0.19743611
## 1403 -0.61121711
## 1404 -0.17155659
## 1406  1.25458114
## 1407  3.95106070
## 1415 -0.61121710
## 1416 -0.61121710
##
## $experiment_day
##      (Intercept)
## 1  8.528852e-10
## 2 -1.965689e-09
## 3  2.091111e-09
## 4 -9.660408e-10
##
## with conditional variances for "mouse_id" "experiment_day"
```

```
VarCorr(ifn_glmr)
```

```
## Groups      Name      Std.Dev.
## mouse_id    (Intercept) 9.0073e-01
## experiment_day (Intercept) 3.1091e-05
## Residual                    7.4498e-01
```

Run emmeans for ifn

```
emm_ifn <- emmeans(ifn_glmr,
  specs = c("t_muris_dose", "lab_or_rw"),
  by = c("treatment"),
  type = "response")
emm_ifn
```

```
## treatment = IgG:
## t_muris_dose lab_or_rw response      SE df asymp.LCL asymp.UCL
## Zero        Lab        1.514   0.590 Inf    0.706    3.25
## High        Lab        0.998   0.476 Inf    0.392    2.54
## Zero        RW         0.994   0.411 Inf    0.442    2.23
## High        RW        11.779   4.590 Inf    5.489   25.28
##
## treatment = CD28:
## t_muris_dose lab_or_rw response      SE df asymp.LCL asymp.UCL
## Zero        Lab        3.089   1.200 Inf    1.439    6.63
## High        Lab       70.935  33.900 Inf   27.840  180.74
## Zero        RW        13.725   5.670 Inf    6.106   30.85
## High        RW       337.694 132.000 Inf  157.351  724.73
##
```

```

## treatment = LPS:
## t_muris_dose lab_or_rw response      SE df asymp.LCL asymp.UCL
## Zero         Lab         1.738   0.677 Inf    0.810    3.73
## High         Lab         1.310   0.625 Inf    0.514    3.34
## Zero         RW          0.903   0.373 Inf    0.402    2.03
## High         RW         21.839   8.510 Inf   10.176   46.87
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale

pairs(emm_ifn)

## treatment = IgG:
## contrast      ratio      SE df null z.ratio p.value
## Zero Lab / High Lab 1.51701 0.93500 Inf 1 0.676 0.9061
## Zero Lab / Zero RW 1.52303 0.86500 Inf 1 0.741 0.8806
## Zero Lab / High RW 0.12855 0.07080 Inf 1 -3.723 0.0011
## High Lab / Zero RW 1.00396 0.63400 Inf 1 0.006 1.0000
## High Lab / High RW 0.08474 0.05220 Inf 1 -4.006 0.0004
## Zero RW / High RW 0.08440 0.04790 Inf 1 -4.353 <0.0001
##
## treatment = CD28:
## contrast      ratio      SE df null z.ratio p.value
## Zero Lab / High Lab 0.04355 0.02680 Inf 1 -5.087 <0.0001
## Zero Lab / Zero RW 0.22508 0.12800 Inf 1 -2.626 0.0429
## Zero Lab / High RW 0.00915 0.00504 Inf 1 -8.519 <0.0001
## High Lab / Zero RW 5.16836 3.26000 Inf 1 2.602 0.0458
## High Lab / High RW 0.21006 0.12900 Inf 1 -2.533 0.0550
## Zero RW / High RW 0.04064 0.02310 Inf 1 -5.639 <0.0001
##
## treatment = LPS:
## contrast      ratio      SE df null z.ratio p.value
## Zero Lab / High Lab 1.32730 0.81800 Inf 1 0.460 0.9678
## Zero Lab / Zero RW 1.92480 1.09000 Inf 1 1.153 0.6566
## Zero Lab / High RW 0.07959 0.04390 Inf 1 -4.593 <0.0001
## High Lab / Zero RW 1.45016 0.91500 Inf 1 0.589 0.9355
## High Lab / High RW 0.05996 0.03690 Inf 1 -4.568 <0.0001
## Zero RW / High RW 0.04135 0.02350 Inf 1 -5.609 <0.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
## Tests are performed on the log scale

# Pairwise contrast: Lab vs Rewilded within each treatment
emm_ifn_env <- emmeans(ifn_glm,
  specs = "t_muris_dose",
  by = c("treatment", "lab_or_rw"),
  type = "response")

pairs(emm_ifn_env)

## treatment = IgG, lab_or_rw = Lab:
## contrast      ratio      SE df null z.ratio p.value
## Zero / High 1.5170 0.9350 Inf 1 0.676 0.4987

```



```
##
## treatment = CD28, lab_or_rw = Lab:
## contrast      ratio      SE df null z.ratio p.value
## Zero / High 0.0435 0.0268 Inf   1  -5.087 <0.0001
##
## treatment = LPS, lab_or_rw = Lab:
## contrast      ratio      SE df null z.ratio p.value
## Zero / High 1.3273 0.8180 Inf   1   0.460  0.6458
##
## treatment = IgG, lab_or_rw = RW:
## contrast      ratio      SE df null z.ratio p.value
## Zero / High 0.0844 0.0479 Inf   1  -4.353 <0.0001
##
## treatment = CD28, lab_or_rw = RW:
## contrast      ratio      SE df null z.ratio p.value
## Zero / High 0.0406 0.0231 Inf   1  -5.639 <0.0001
##
## treatment = LPS, lab_or_rw = RW:
## contrast      ratio      SE df null z.ratio p.value
## Zero / High 0.0413 0.0235 Inf   1  -5.609 <0.0001
##
## Tests are performed on the log scale
```

Doing the same thing here

The lowest detection level is 3.7 for TNF so halving that.

```
tnf_input <- tnf_input %>%
  mutate(
    t_muris_dose = relevel(factor(t_muris_dose),
                              ref = "Zero"),
    treatment = relevel(factor(treatment),
                          ref = "IgG"),
    lab_or_rw = relevel(factor(lab_or_rw),
                          ref = "Lab")
  )

tnf_glmer_input <- tnf_input %>%
  mutate(mean_concentration = ifelse(mean_concentration == 0,
                                     3.7 / 2,
                                     mean_concentration))

# we want to set the zero dose, IgG and lab groups to be the basis of all our comparisons.

tnf_glmer <- glmer(mean_concentration ~ lab_or_rw *
                  t_muris_dose *
                  treatment +
                  (1 | mouse_id) +
                  (1 | experiment_day),
                  family = Gamma(link = "log"),
                  control = glmerControl(optimizer = "bobyqa"),
```

```

                                optCtrl = list(maxfun = 2e5)),
data = tnf_glmer_input)

```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(tnf_glmer)
```

```

## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##   Family: Gamma ( log )
## Formula: mean_concentration ~ lab_or_rw * t_muris_dose * treatment + (1 |
##   mouse_id) + (1 | experiment_day)
##   Data: tnf_glmer_input
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##           AIC          BIC      logLik -2*log(L)  df.resid
##       952.7       991.2    -461.3     922.7       81
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.21568 -0.30914  0.07227  0.53153  1.82232
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
##  mouse_id      (Intercept)  0.1011     0.3180
##  experiment_day (Intercept)  0.0000     0.0000
##  Residual                0.1854     0.4306
## Number of obs: 96, groups:  mouse_id, 32; experiment_day, 4
##
## Fixed effects:
##
##              Estimate Std. Error t value Pr(>|z|)
## (Intercept)      3.13748    0.17843  17.584 < 2e-16
## lab_or_rwRW     -0.05402    0.26011  -0.208  0.835464
## t_muris_doseHigh  0.61238    0.28213   2.171  0.029963
## treatmentCD28     0.20566    0.20299   1.013  0.310977
## treatmentLPS      0.78482    0.20299   3.866  0.000111
## lab_or_rwRW:t_muris_doseHigh  0.60202    0.38373   1.569  0.116681
## lab_or_rwRW:treatmentCD28    0.45128    0.29591   1.525  0.127241
## lab_or_rwRW:treatmentLPS     0.12976    0.29591   0.439  0.661011
## t_muris_doseHigh:treatmentCD28 0.17624    0.32095   0.549  0.582918
## t_muris_doseHigh:treatmentLPS -0.44592    0.32095  -1.389  0.164726
## lab_or_rwRW:t_muris_doseHigh:treatmentCD28 -0.32167    0.43655  -0.737  0.461204
## lab_or_rwRW:t_muris_doseHigh:treatmentLPS  0.14191    0.43655   0.325  0.745120
##
## (Intercept)          ***
## lab_or_rwRW
## t_muris_doseHigh      *
## treatmentCD28
## treatmentLPS          ***
## lab_or_rwRW:t_muris_doseHigh
## lab_or_rwRW:treatmentCD28
## lab_or_rwRW:treatmentLPS

```

```

## t_muris_doseHigh:treatmentCD28
## t_muris_doseHigh:treatmentLPS
## lab_or_rwRW:t_muris_doseHigh:treatmentCD28
## lab_or_rwRW:t_muris_doseHigh:treatmentLPS
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) lb__RW t_mr_H trCD28 trtLPS lb__RW:__H l__RW:C l__RW:L
## lab_or_rwRW -0.686
## t_mrs_dsHgh -0.632  0.434
## tretmntCD28 -0.569  0.390  0.360
## treatmntLPS -0.569  0.390  0.360  0.500
## lb_r_RW:__H  0.465 -0.678 -0.735 -0.264 -0.264
## lb__RW:CD28  0.390 -0.569 -0.247 -0.686 -0.343  0.386
## lb_r_RW:LPS  0.390 -0.569 -0.247 -0.343 -0.686  0.386      0.500
## t_mr_H:CD28  0.360 -0.247 -0.569 -0.632 -0.316  0.418      0.434  0.217
## t_mrs_H:LPS  0.360 -0.247 -0.569 -0.316 -0.632  0.418      0.217  0.434
## l__RW:__H:C -0.264  0.386  0.418  0.465  0.232 -0.569      -0.678 -0.339
## l__RW:__H:L -0.264  0.386  0.418  0.232  0.465 -0.569      -0.339 -0.678
##      t__H:C t__H:L l__RW:__H:C
## lab_or_rwRW
## t_mrs_dsHgh
## tretmntCD28
## treatmntLPS
## lb_r_RW:__H
## lb__RW:CD28
## lb_r_RW:LPS
## t_mr_H:CD28
## t_mrs_H:LPS  0.500
## l__RW:__H:C -0.735 -0.368
## l__RW:__H:L -0.368 -0.735  0.500
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

```

Diagnostics for tn timer

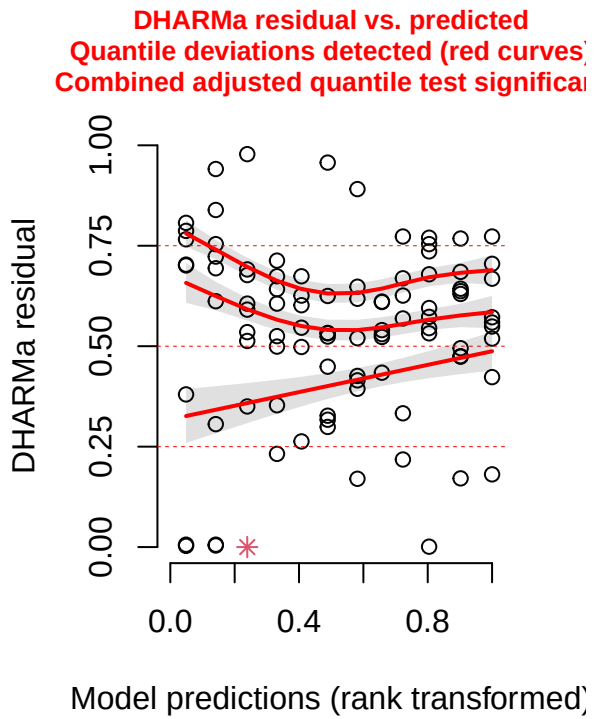
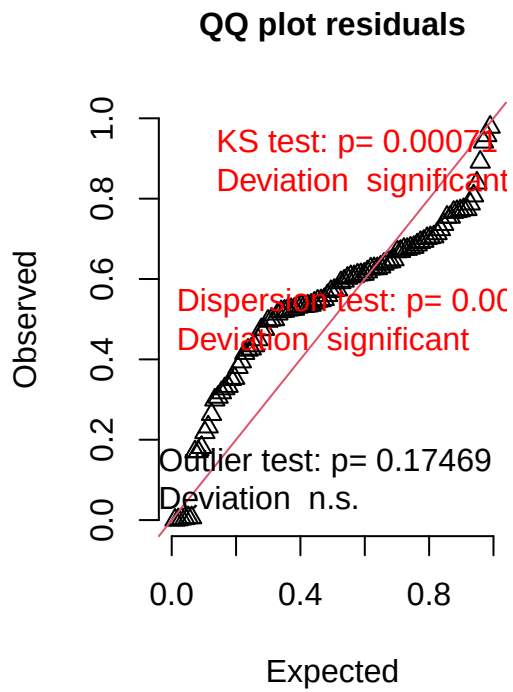
```

tnf_sim_res <- simulateResiduals(fittedModel = tnf_glmer, n = 1000)

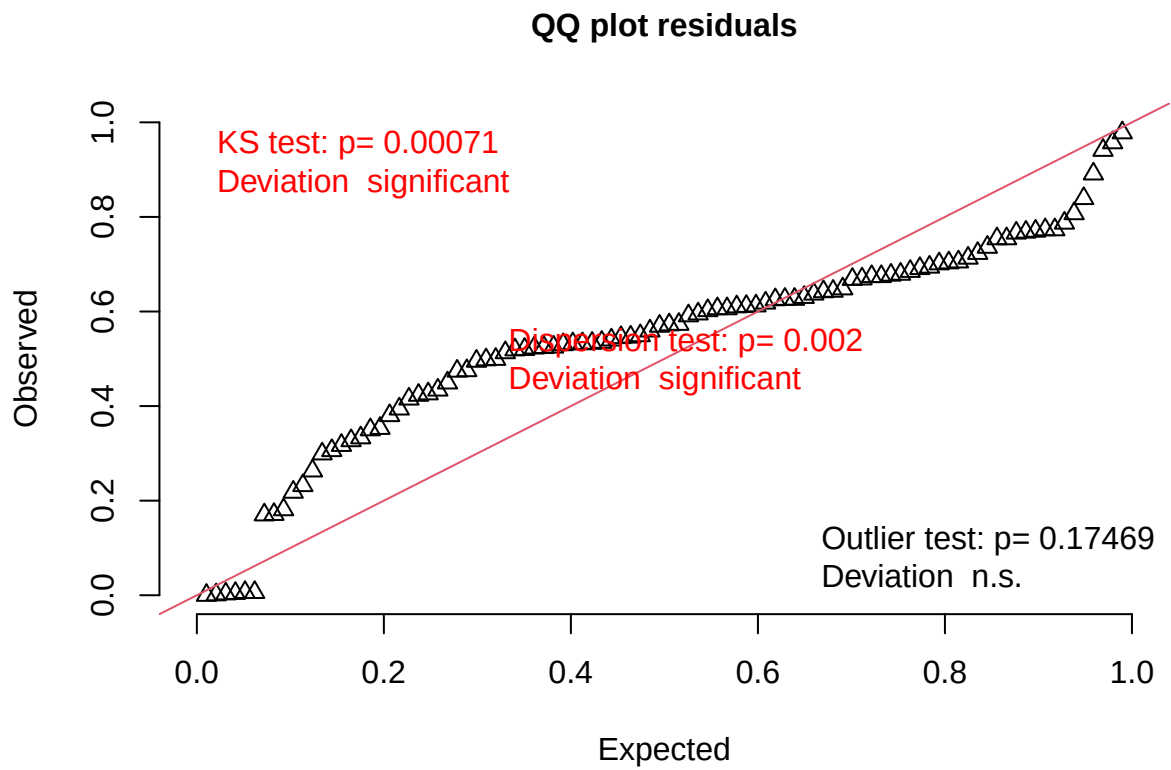
plot(tnf_sim_res)

```

DHARMa residual



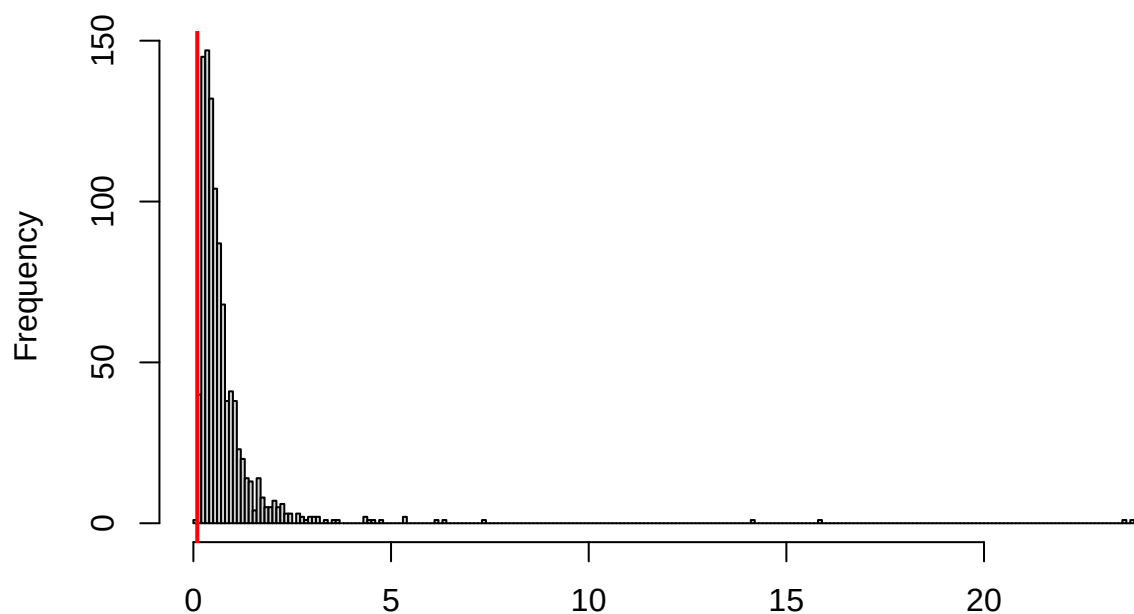
```
testUniformity(tnf_sim_res)
```



```
##  
## Asymptotic one-sample Kolmogorov-Smirnov test  
##  
## data: simulationOutput$scaledResiduals  
## D = 0.20333, p-value = 0.0007137  
## alternative hypothesis: two-sided
```

```
testDispersion(tnf_sim_res)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**

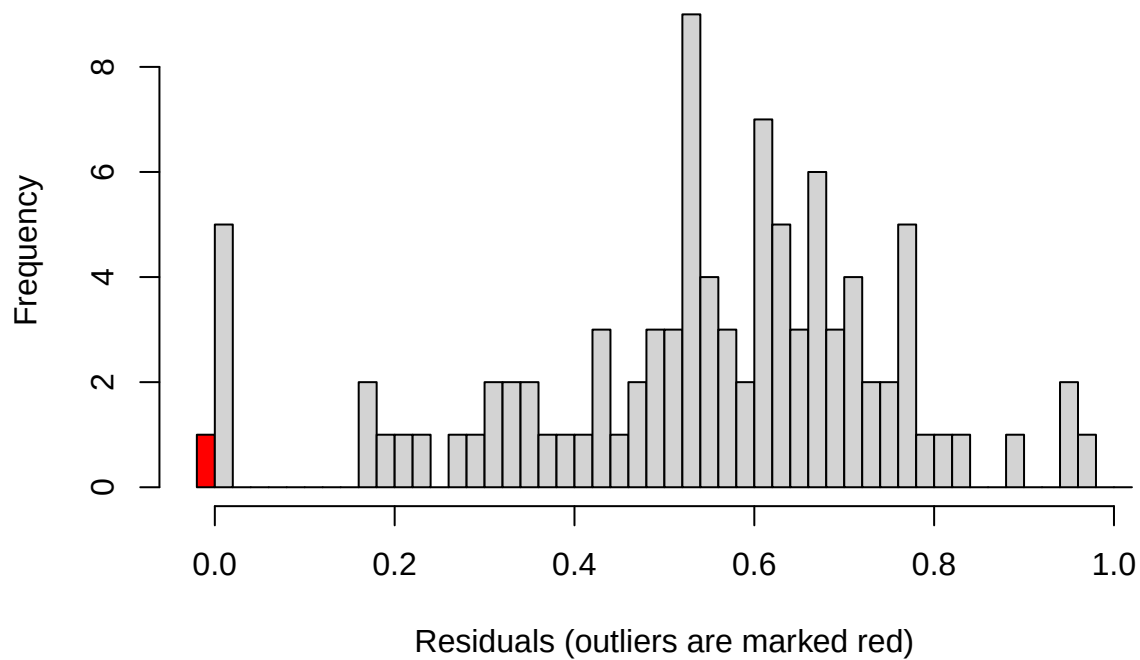


Simulated values, red line = fitted model. p-value (two.sided) = 0.002

```
##  
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.  
## simulated  
##  
## data: simulationOutput  
## dispersion = 0.12229, p-value = 0.002  
## alternative hypothesis: two.sided
```

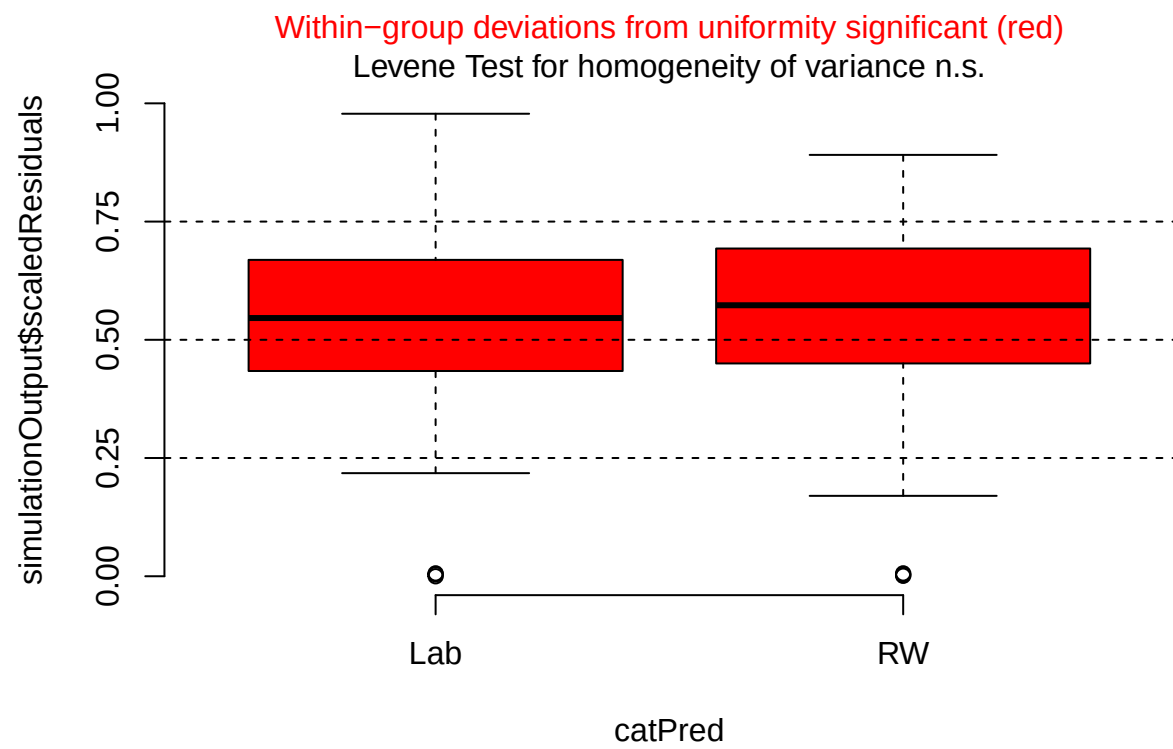
```
testOutliers(tnf_sim_res)
```

Outlier test n.s.

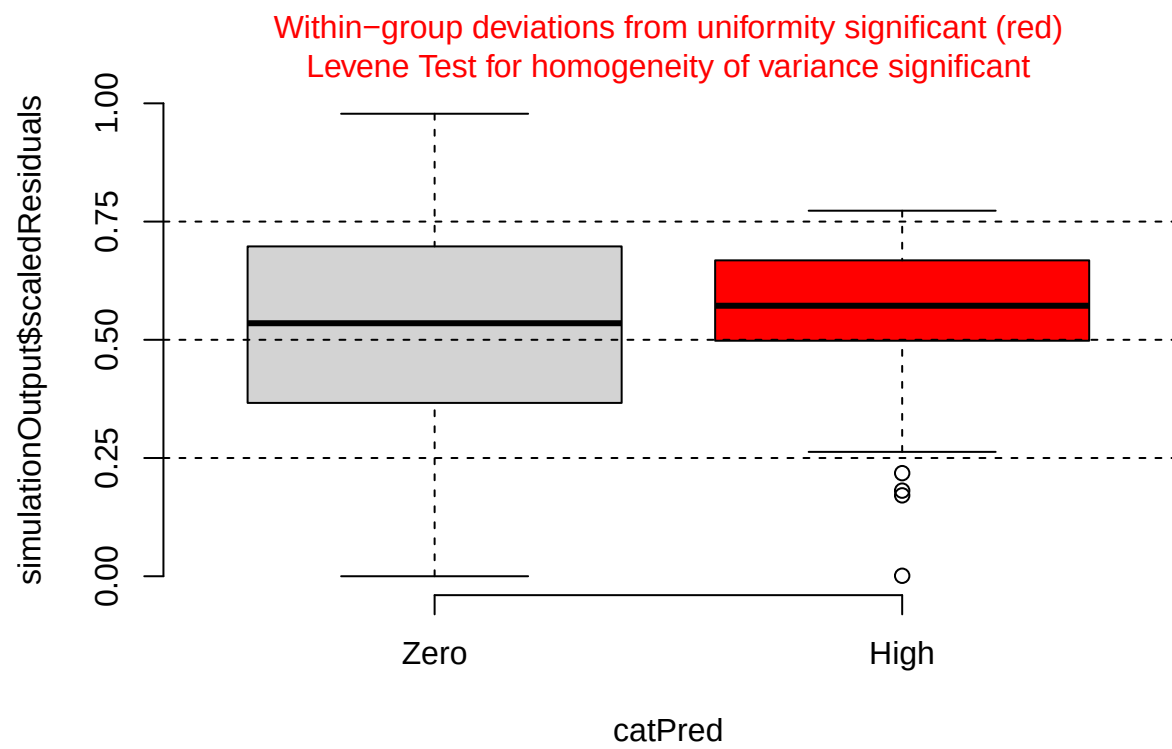


```
##
## DHARMA outlier test based on exact binomial test with approximate
## expectations
##
## data:  tnf_sim_res
## outliers at both margin(s) = 1, observations = 96, p-value = 0.1747
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
##  0.0002636924 0.0566747070
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                0.01041667
```

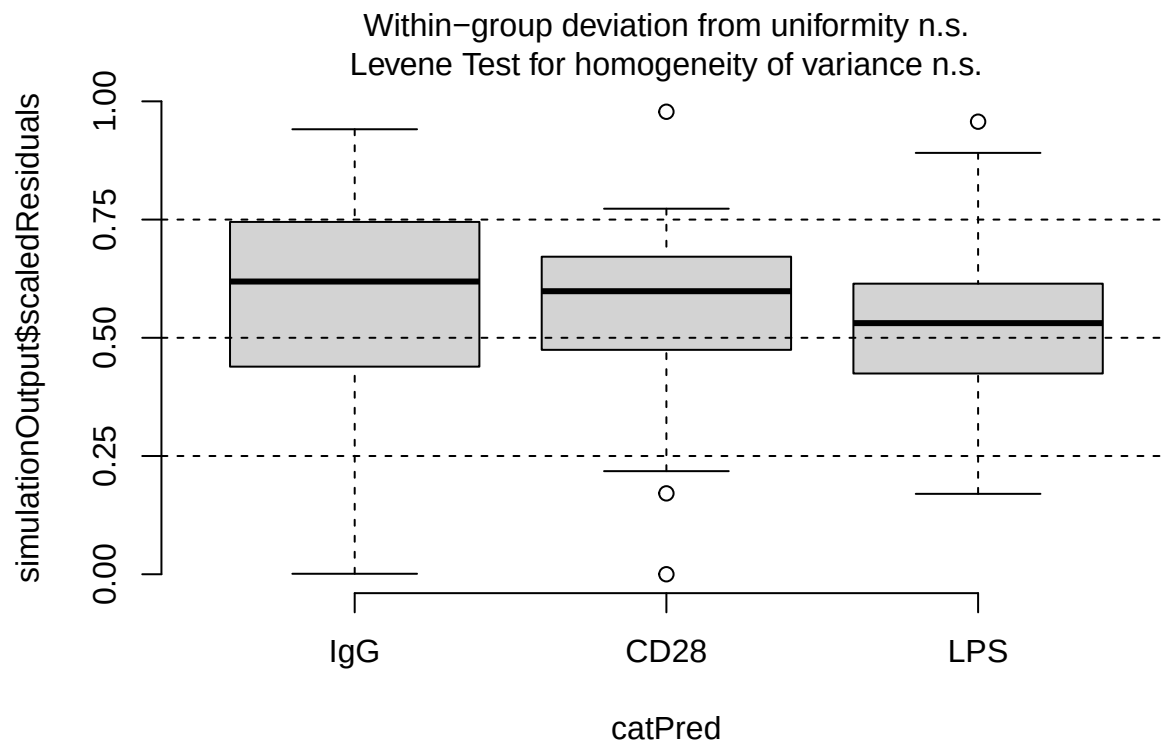
```
plotResiduals(tnf_sim_res, form = tn timer_glm er_input$lab_or_rw)
```



```
plotResiduals(tnf_sim_res, form = tnf_glmer_input$t_muris_dose)
```

```
plotResiduals(tnf_sim_res, form = tn timer_glm er_input$treatment)
```



```
ranef(tnf_glmmer)
```

```
## $mouse_id
##      (Intercept)
## 1357 0.042478440
## 1359 0.203210997
## 1360 0.035079580
## 1361 0.194792017
## 1363 0.154857138
## 1364 0.160186038
## 1366 -0.171686110
## 1367 0.133837470
## 1368 0.061095941
## 1370 0.145734526
## 1371 0.191508907
## 1373 0.044236847
## 1374 0.045002963
## 1376 0.252290298
## 1378 -0.088805877
## 1383 0.090264286
## 1385 -0.400328289
## 1388 -0.007058363
## 1389 -0.086583642
## 1391 -0.726320935
## 1392 0.452596243
```

```
## 1394 0.088506359
## 1395 -0.079435426
## 1396 -0.013217260
## 1398 -0.145309308
## 1401 0.299328038
## 1403 -0.673959683
## 1404 -0.375888380
## 1406 0.257734685
## 1407 0.947138010
## 1415 0.105477764
## 1416 -0.467293747
##
## $experiment_day
## (Intercept)
## 1 0
## 2 0
## 3 0
## 4 0
##
## with conditional variances for "mouse_id" "experiment_day"
```

```
VarCorr(tnf_glmmer)
```

```
## Groups Name Std.Dev.
## mouse_id (Intercept) 0.31800
## experiment_day (Intercept) 0.00000
## Residual 0.43061
```

Run emmeans for tnf

```
emm_tnf <- emmeans(tnf_glmmer,
  specs = c("t_muris_dose", "lab_or_rw"),
  by = c("treatment"),
  type = "response")
```

```
emm_tnf
```

```
## treatment = IgG:
## t_muris_dose lab_or_rw response SE df asymp.LCL asymp.UCL
## Zero Lab 23.0 4.11 Inf 16.2 32.7
## High Lab 42.5 9.29 Inf 27.7 65.2
## Zero RW 21.8 4.13 Inf 15.1 31.6
## High RW 73.5 13.10 Inf 51.8 104.3
##
## treatment = CD28:
## t_muris_dose lab_or_rw response SE df asymp.LCL asymp.UCL
## Zero Lab 28.3 5.05 Inf 20.0 40.2
## High Lab 62.3 13.60 Inf 40.6 95.6
## Zero RW 42.1 7.97 Inf 29.1 61.0
## High RW 122.7 21.90 Inf 86.5 174.0
```

```
##
## treatment = LPS:
## t_muris_dose lab_or_rw response SE df asymp.LCL asymp.UCL
## Zero Lab 50.5 9.01 Inf 35.6 71.7
## High Lab 59.7 13.00 Inf 38.9 91.6
## Zero RW 54.5 10.30 Inf 37.6 79.0
## High RW 135.4 24.20 Inf 95.5 192.1
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale

pairs(emm_tnf)

## treatment = IgG:
## contrast ratio SE df null z.ratio p.value
## Zero Lab / High Lab 0.542 0.1530 Inf 1 -2.171 0.1315
## Zero Lab / Zero RW 1.056 0.2750 Inf 1 0.208 0.9968
## Zero Lab / High RW 0.313 0.0791 Inf 1 -4.598 <0.0001
## High Lab / Zero RW 1.947 0.5630 Inf 1 2.305 0.0968
## High Lab / High RW 0.578 0.1630 Inf 1 -1.942 0.2104
## Zero RW / High RW 0.297 0.0772 Inf 1 -4.669 <0.0001
##
## treatment = CD28:
## contrast ratio SE df null z.ratio p.value
## Zero Lab / High Lab 0.454 0.1280 Inf 1 -2.795 0.0266
## Zero Lab / Zero RW 0.672 0.1750 Inf 1 -1.527 0.4211
## Zero Lab / High RW 0.231 0.0582 Inf 1 -5.810 <0.0001
## High Lab / Zero RW 1.479 0.4280 Inf 1 1.354 0.5286
## High Lab / High RW 0.508 0.1430 Inf 1 -2.402 0.0767
## Zero RW / High RW 0.343 0.0893 Inf 1 -4.110 0.0002
##
## treatment = LPS:
## contrast ratio SE df null z.ratio p.value
## Zero Lab / High Lab 0.847 0.2390 Inf 1 -0.590 0.9351
## Zero Lab / Zero RW 0.927 0.2410 Inf 1 -0.291 0.9914
## Zero Lab / High RW 0.373 0.0941 Inf 1 -3.908 0.0005
## High Lab / Zero RW 1.095 0.3170 Inf 1 0.314 0.9893
## High Lab / High RW 0.441 0.1240 Inf 1 -2.905 0.0192
## Zero RW / High RW 0.402 0.1050 Inf 1 -3.500 0.0026
##
## P value adjustment: tukey method for comparing a family of 4 estimates
## Tests are performed on the log scale

# Pairwise contrast: Lab vs Rewilded within each treatment
emm_tnf_env <- emmeans(tnf_glmr,
  specs = "t_muris_dose",
  by = c("treatment", "lab_or_rw"),
  type = "response")

pairs(emm_tnf_env)

## treatment = IgG, lab_or_rw = Lab:
## contrast ratio SE df null z.ratio p.value
```

```
## Zero / High 0.542 0.1530 Inf    1 -2.171  0.0300
##
## treatment = CD28, lab_or_rw = Lab:
## contrast      ratio      SE  df null z.ratio p.value
## Zero / High 0.454 0.1280 Inf    1 -2.795  0.0052
##
## treatment = LPS, lab_or_rw = Lab:
## contrast      ratio      SE  df null z.ratio p.value
## Zero / High 0.847 0.2390 Inf    1 -0.590  0.5552
##
## treatment = IgG, lab_or_rw = RW:
## contrast      ratio      SE  df null z.ratio p.value
## Zero / High 0.297 0.0772 Inf    1 -4.669 <0.0001
##
## treatment = CD28, lab_or_rw = RW:
## contrast      ratio      SE  df null z.ratio p.value
## Zero / High 0.343 0.0893 Inf    1 -4.110 <0.0001
##
## treatment = LPS, lab_or_rw = RW:
## contrast      ratio      SE  df null z.ratio p.value
## Zero / High 0.402 0.1050 Inf    1 -3.500  0.0005
##
## Tests are performed on the log scale
```

EMM data frames

```
emm_ifn_df <- as.data.frame(emm_ifn) %>%
  mutate(treatment = factor(treatment, levels = c("IgG", "LPS", "CD28")),
         t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High")),
         lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")))

emm_tnf_df <- as.data.frame(emm_tnf) %>%
  mutate(treatment = factor(treatment, levels = c("IgG", "LPS", "CD28")),
         t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High")),
         lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")))
```

EMM IFN-gamma: #1 — Facet = Treatment, x = Environment, Color = Dose

“Within each treatment, how does infection shift the response across environments?”

```
# We're plotting the LOD/2 values as the data points as well.
# That is what those clustered points are. Those were zeros.

emm_ifn_plot <- ggplot(emm_ifn_df) +
  geom_point(data = ifn_glmer_input %>%
```

```

mutate(treatment = factor(treatment,
                           levels = c("IgG", "LPS", "CD28")),
       lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")),
       t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High"))),
aes(x = lab_or_rw,
    y = mean_concentration,
    color = t_muris_dose),
position = position_jitterdodge(jitter.width = 0.1,
                                dodge.width = 0.5),

alpha = 0.3,
size = 1.5) +
geom_errorbar(aes(x = lab_or_rw,
                  ymin = asymp.LCL,
                  ymax = asymp.UCL,
                  color = t_muris_dose),
              width = 0.3,
              position = position_dodge(width = 0.5),
              alpha = 0.8) +
geom_point(aes(x = lab_or_rw,
               y = response,
               fill = t_muris_dose),
           position = position_dodge(width = 0.5),
           alpha = 1,
           size = 3,
           shape = 21,
           color = "black") +
scale_fill_manual(values = c("Zero" = color_dark_blue,
                             "High" = color_orange),
                  name = expression(bolditalic("T. muris") ~ bold("Dose")) +
scale_color_manual(values = c("Zero" = color_dark_blue,
                              "High" = color_orange),
                   name = expression(bolditalic("T. muris") ~ bold("Dose")) +
scale_y_continuous(name = expression(bold("IFN-" * gamma ~
                                     "(pg mL"^-1" * ")")),
                  expand = expansion(mult = 0),
                  trans = "log10",
                  breaks = 10^seq(-1, 4),
                  labels = expression(10^-1, 1, 10^1,
                                     10^2, 10^3, 10^4),
                  limits = c(0.1, 10000),
                  guide = guide_axis_logticks(long = 1.8,
                                               mid = 1.3,
                                               short = 0.8,
                                               expanded = FALSE)) +
scale_x_discrete(name = expression(bold("Environment")),
                 labels = c("Lab" = "Laboratory",
                           "RW" = "Rewilded")) +
facet_wrap(~ treatment,
           labeller = as_labeller(c("CD28" = "CD28SA",
                                    "LPS" = "LPS",
                                    "IgG" = "IgG1")) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",

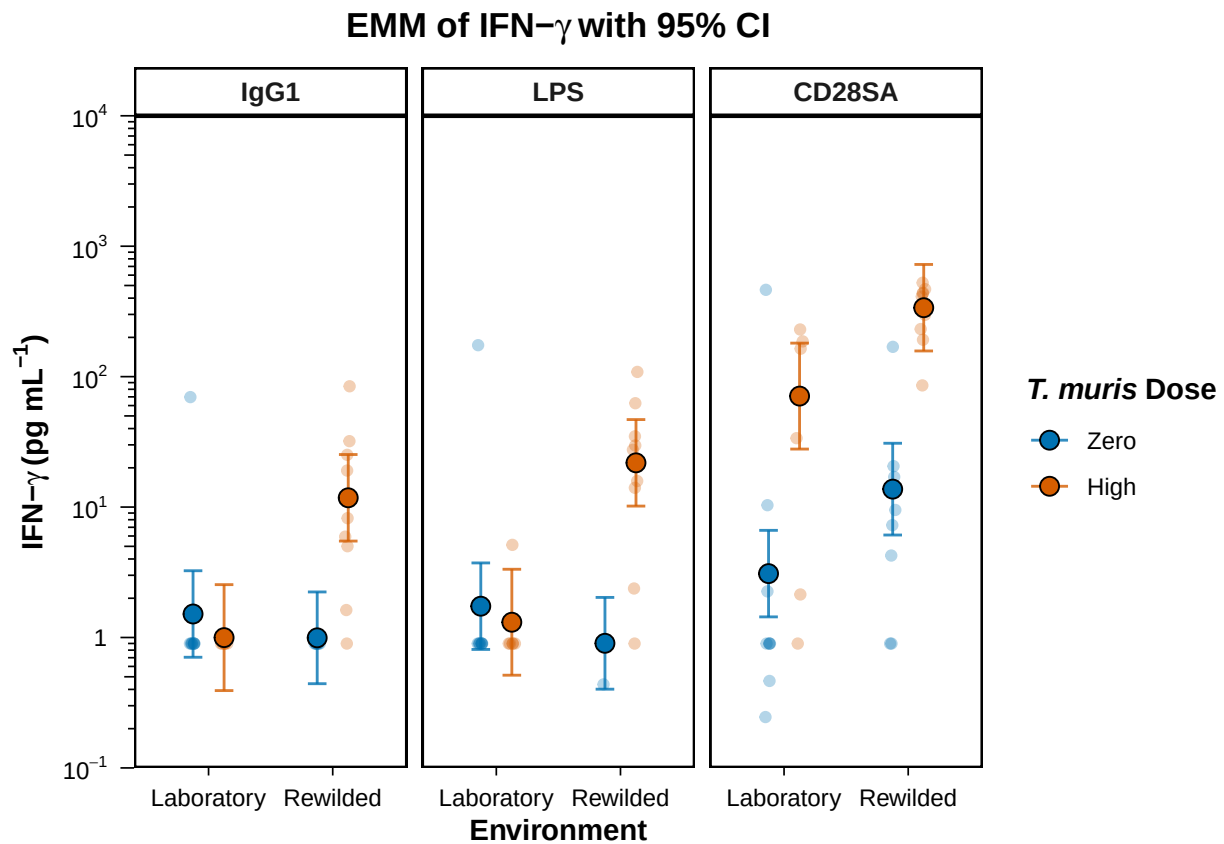
```

```

        color = "black",
        linewidth = 0.8),
strip.text = element_text(size = 10,
                           face = "bold"),
panel.border = element_rect(color = "black",
                             fill = NA,
                             linewidth = 0.8),
panel.grid = element_blank(),
plot.title = element_text(hjust = 0.5, face = "bold"),
axis.title.x = element_text(face = "bold"),
axis.title.y = element_text(face = "bold"),
axis.text = element_text(color = "black"),
axis.ticks = element_line(color = "black",
                           linewidth = 0.3),
axis.ticks.length = unit(0.1, "cm")) +
ggtitle(expression(bold("EMM of IFN- $\gamma$  * gamma ~ "with 95% CI"))))

```

emm_ifn_plot



```

plot_saver("emm_ifn_plot.png",
           emm_ifn_plot)

```

EMM IFN-gamma: #5 — Facet = Dose, x = Treatment, Color = Environment

“Among infected (or uninfected) mice, which treatment shows the biggest environment effect?”

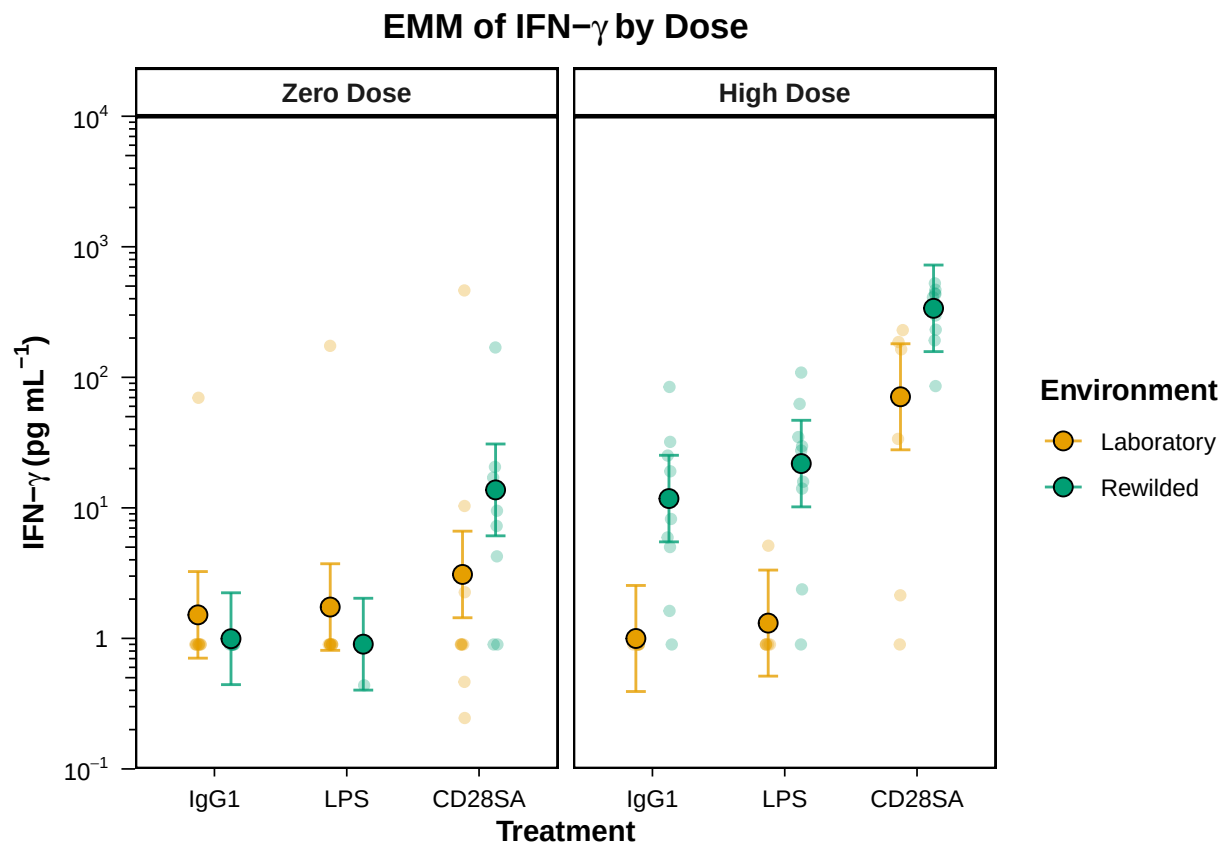
```
emm_ifn_dose_plot <- ggplot(emm_ifn_df) +  
  geom_point(data = ifn_glmer_input %>%  
    mutate(treatment = factor(treatment,  
                              levels = c("IgG", "LPS", "CD28")),  
           lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")),  
           t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High"))),  
    aes(x = treatment,  
        y = mean_concentration,  
        color = lab_or_rw),  
    position = position_jitterdodge(jitter.width = 0.1,  
                                    dodge.width = 0.5),  
    alpha = 0.3,  
    size = 1.5) +  
  geom_errorbar(aes(x = treatment,  
                   ymin = asymp.LCL,  
                   ymax = asymp.UCL,  
                   color = lab_or_rw),  
               width = 0.3,  
               position = position_dodge(width = 0.5),  
               alpha = 0.8) +  
  geom_point(aes(x = treatment,  
                 y = response,  
                 fill = lab_or_rw),  
             position = position_dodge(width = 0.5),  
             alpha = 1,  
             size = 3,  
             shape = 21,  
             color = "black") +  
  scale_fill_manual(values = c("Lab" = color_mustard,  
                               "RW" = color_green),  
                   name = expression(bold("Environment")),  
                   labels = c("Lab" = "Laboratory",  
                              "RW" = "Rewilded")) +  
  scale_color_manual(values = c("Lab" = color_mustard,  
                                "RW" = color_green),  
                    name = expression(bold("Environment")),  
                    labels = c("Lab" = "Laboratory",  
                               "RW" = "Rewilded")) +  
  scale_y_continuous(name = expression(bold("IFN-" * gamma ~  
                                       "(pg mL"^-1 * ")")),  
                    expand = expansion(mult = 0),  
                    trans = "log10",  
                    breaks = 10^seq(-1, 4),  
                    labels = expression(10^-1, 1, 10^1,  
                                         10^2, 10^3, 10^4),
```



```

        limits = c(0.1, 10000),
        guide = guide_axis_logticks(long = 1.8,
                                    mid = 1.3,
                                    short = 0.8,
                                    expanded = FALSE)) +
scale_x_discrete(name = expression(bold("Treatment")),
                 labels = c("CD28" = "CD28SA",
                           "LPS" = "LPS",
                           "IgG" = "IgG1")) +
facet_wrap(~ t_muris_dose,
           labeller = as_labeller(c("Zero" = "Zero Dose",
                                    "High" = "High Dose")))) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                       color = "black",
                                       linewidth = 0.8),
      strip.text = element_text(size = 10,
                                face = "bold"),
      panel.border = element_rect(color = "black",
                                  fill = NA,
                                  linewidth = 0.8),
      panel.grid = element_blank(),
      plot.title = element_text(hjust = 0.5, face = "bold"),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      axis.text = element_text(color = "black"),
      axis.ticks = element_line(color = "black",
                                linewidth = 0.3),
      axis.ticks.length = unit(0.1, "cm")) +
ggtitle(expression(bold("EMM of IFN- $\gamma$  ~ "by Dose"))))
emm_ifn_dose_plot

```



```
plot_saver("emm_ifn_dose_plot.png",
  emm_ifn_dose_plot)
```

EMM IFN-gamma: #3 — Facet = Environment, x = Treatment, Color = Dose

“Within Lab (or RW), which treatment shows the strongest infection effect?”

```
emm_ifn_treatment_plot <- ggplot(emm_ifn_df) +
  geom_point(data = ifn_glmer_input %>%
    mutate(treatment = factor(treatment,
                              levels = c("IgG", "LPS", "CD28")),
           lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")),
           t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High"))),
    aes(x = treatment,
        y = mean_concentration,
        color = t_muris_dose),
    position = position_jitterdodge(jitter.width = 0.1,
                                    dodge.width = 0.5),
    alpha = 0.3,
```

```

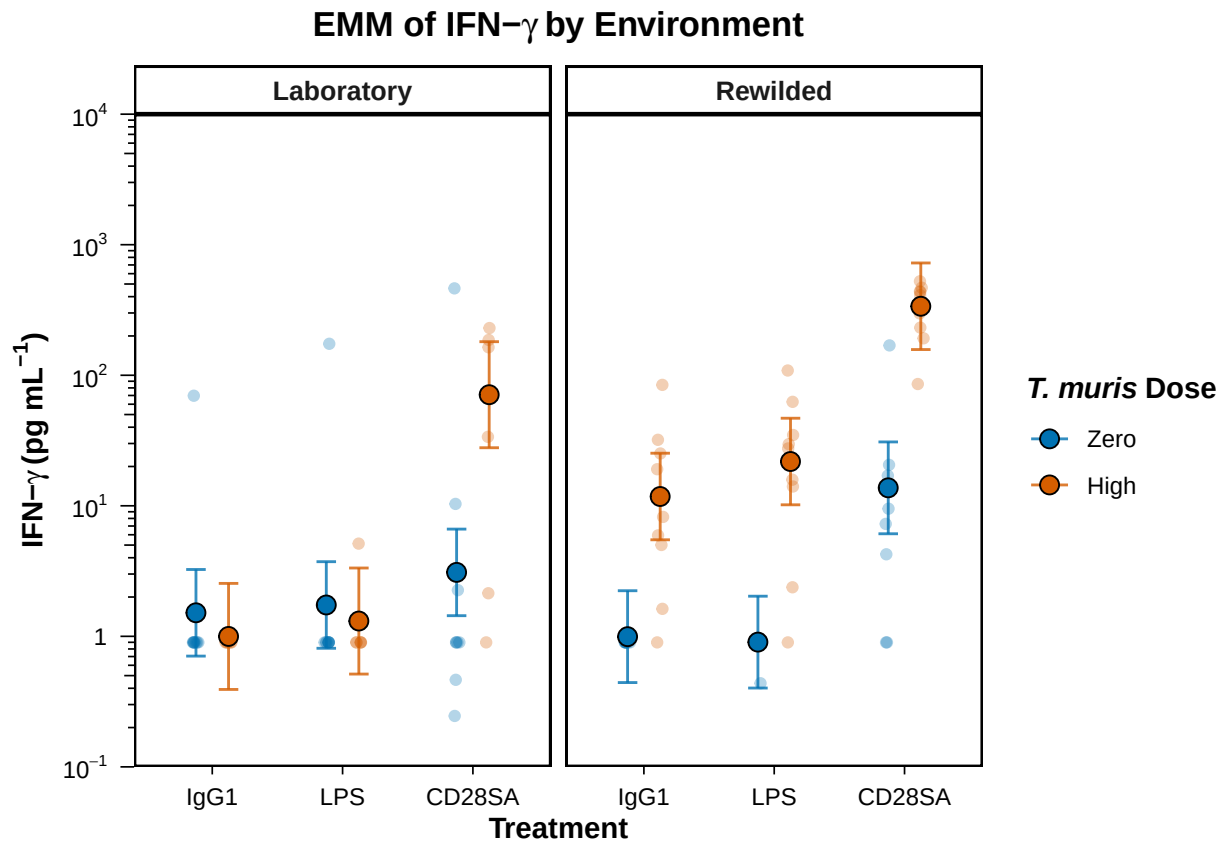
      size = 1.5) +
geom_errorbar(aes(x = treatment,
  ymin = asymp.LCL,
  ymax = asymp.UCL,
  color = t_muris_dose),
  width = 0.3,
  position = position_dodge(width = 0.5),
  alpha = 0.8) +
geom_point(aes(x = treatment,
  y = response,
  fill = t_muris_dose),
  position = position_dodge(width = 0.5),
  alpha = 1,
  size = 3,
  shape = 21,
  color = "black") +
scale_fill_manual(values = c("Zero" = color_dark_blue,
  "High" = color_orange),
  name = expression(bolditalic("T. muris") ~ bold("Dose")) +
scale_color_manual(values = c("Zero" = color_dark_blue,
  "High" = color_orange),
  name = expression(bolditalic("T. muris") ~ bold("Dose")) +
scale_y_continuous(name = expression(bold("IFN-" * gamma ~
  "(pg mL"^-1" * ")")),
  expand = expansion(mult = 0),
  trans = "log10",
  breaks = 10^seq(-1, 4),
  labels = expression(10^-1, 1, 10^1,
  10^2, 10^3, 10^4),
  limits = c(0.1, 10000),
  guide = guide_axis_logticks(long = 1.8,
  mid = 1.3,
  short = 0.8,
  expanded = FALSE)) +
scale_x_discrete(name = expression(bold("Treatment")),
  labels = c("CD28" = "CD28SA",
  "LPS" = "LPS",
  "IgG" = "IgG1")) +
facet_wrap(~ lab_or_rw,
  labeller = as_labeller(c("Lab" = "Laboratory",
  "RW" = "Rewilded")) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
  color = "black",
  linewidth = 0.8),
  strip.text = element_text(size = 10,
  face = "bold"),
  panel.border = element_rect(color = "black",
  fill = NA,
  linewidth = 0.8),
  panel.grid = element_blank(),
  plot.title = element_text(hjust = 0.5, face = "bold"),
  axis.title.x = element_text(face = "bold"),

```

```
axis.title.y = element_text(face = "bold"),
axis.text = element_text(color = "black"),
axis.ticks = element_line(color = "black",
                           linewidth = 0.3),

axis.ticks.length = unit(0.1, "cm")) +
ggtitle(expression(bold("EMM of IFN- $\gamma$  * gamma ~ "by Environment")))
```

emm_ifn_treatment_plot



```
plot_saver("emm_ifn_treatment_plot.png",
            emm_ifn_treatment_plot)
```

EMM TNF-alpha: #1 — Facet = Treatment, x = Environment, Color = Dose

```
emm_tnf_plot <- ggplot(emm_tnf_df) +
  geom_point(data = tnf_glmmer_input %>%
    mutate(treatment = factor(treatment,
                              levels = c("IgG", "LPS", "CD28")),
           lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")),
           t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High"))),
```

```

aes(x = lab_or_rw,
    y = mean_concentration,
    color = t_muris_dose),
position = position_jitterdodge(jitter.width = 0.1,
                                dodge.width = 0.5),

alpha = 0.3,
size = 1.5) +
geom_errorbar(aes(x = lab_or_rw,
    ymin = asymp.LCL,
    ymax = asymp.UCL,
    color = t_muris_dose),
    width = 0.3,
    position = position_dodge(width = 0.5),
    alpha = 0.8) +
geom_point(aes(x = lab_or_rw,
    y = response,
    fill = t_muris_dose),
    position = position_dodge(width = 0.5),
    alpha = 1,
    size = 3,
    shape = 21,
    color = "black") +
scale_fill_manual(values = c("Zero" = color_dark_blue,
    "High" = color_orange),
    name = expression(bolditalic("T. muris") ~ bold("Dose")) +
scale_color_manual(values = c("Zero" = color_dark_blue,
    "High" = color_orange),
    name = expression(bolditalic("T. muris") ~ bold("Dose")) +
scale_y_continuous(name = expression(bold("TNF-" * alpha ~
    "(pg mL"^-1" * ")")),
    expand = expansion(mult = 0),
    trans = "log10",
    breaks = 10^seq(-1, 4),
    labels = expression(10^-1, 1, 10^1,
    10^2, 10^3, 10^4),
    limits = c(0.1, 10000),
    guide = guide_axis_logticks(long = 1.8,
    mid = 1.3,
    short = 0.8,
    expanded = FALSE)) +
scale_x_discrete(name = expression(bold("Environment")),
    labels = c("Lab" = "Laboratory",
    "RW" = "Rewilded")) +
facet_wrap(~ treatment,
    labeller = as_labeller(c("CD28" = "CD28SA",
    "LPS" = "LPS",
    "IgG" = "IgG1")) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
    color = "black",
    linewidth = 0.8),
    strip.text = element_text(size = 10,
    face = "bold"),

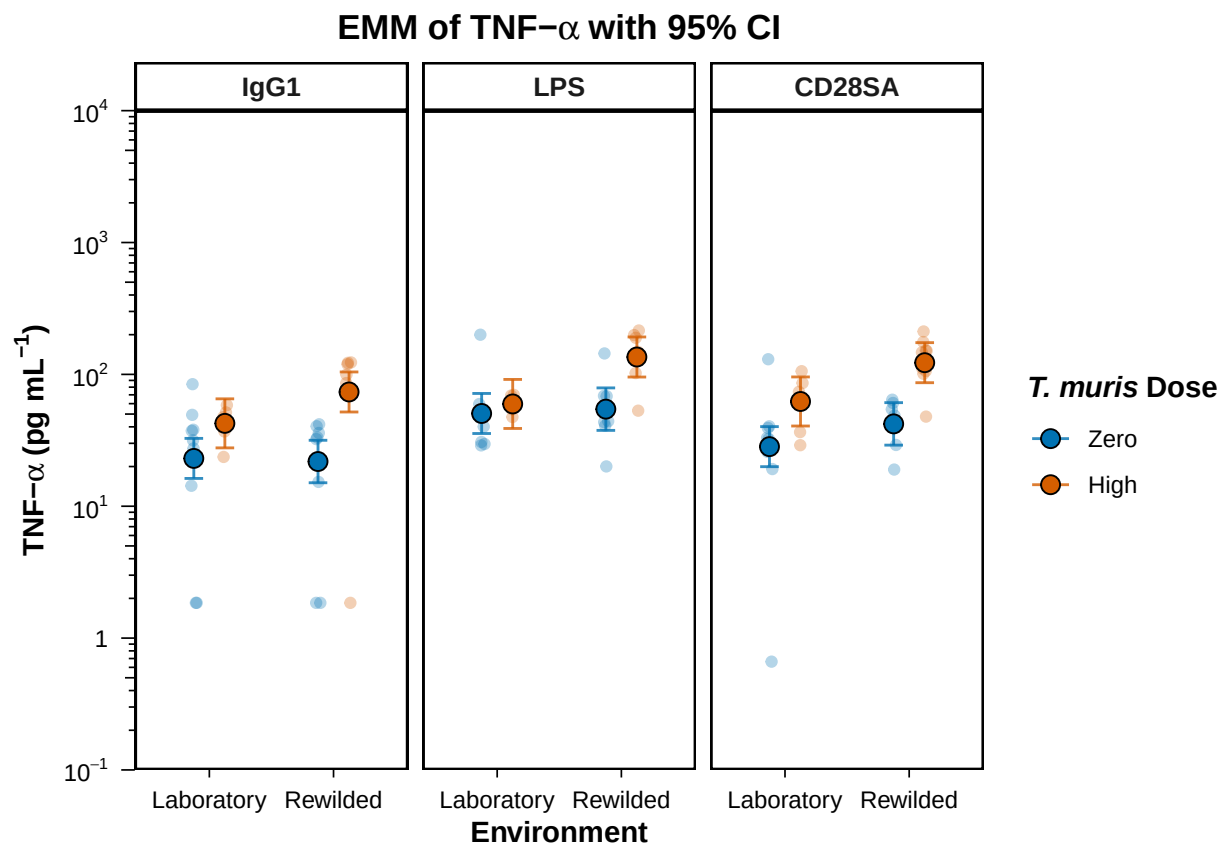
```

```

panel.border = element_rect(color = "black",
                             fill = NA,
                             linewidth = 0.8),
panel.grid = element_blank(),
plot.title = element_text(hjust = 0.5, face = "bold"),
axis.title.x = element_text(face = "bold"),
axis.title.y = element_text(face = "bold"),
axis.text = element_text(color = "black"),
axis.ticks = element_line(color = "black",
                           linewidth = 0.3),
axis.ticks.length = unit(0.1, "cm")) +
ggtitle(expression(bold("EMM of TNF- $\alpha$  * alpha ~ "with 95% CI"))))

```

emm_tnf_plot



```

plot_saver("emm_tnf_plot.png",
            emm_tnf_plot)

```

EMM TNF- α : #5 — Facet = Dose, x = Treatment, Color = Environment

```

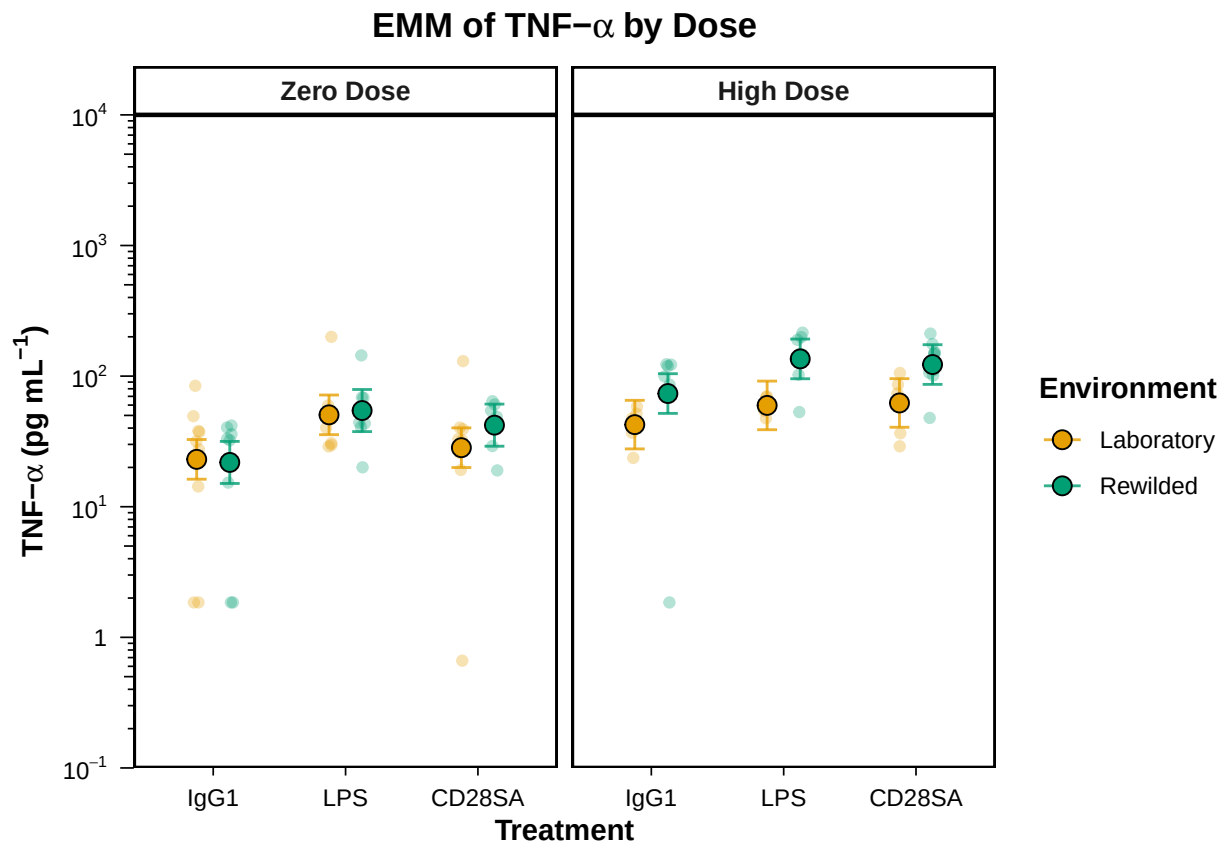
emm_tnf_dose_plot <- ggplot(emm_tnf_df) +
  geom_point(data = tnf_glmer_input %>%
    mutate(treatment = factor(treatment,
                              levels = c("IgG", "LPS", "CD28")),
           lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")),
           t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High"))),
    aes(x = treatment,
         y = mean_concentration,
         color = lab_or_rw),
    position = position_jitterdodge(jitter.width = 0.1,
                                    dodge.width = 0.5),
    alpha = 0.3,
    size = 1.5) +
  geom_errorbar(aes(x = treatment,
                    ymin = asymp.LCL,
                    ymax = asymp.UCL,
                    color = lab_or_rw),
    width = 0.3,
    position = position_dodge(width = 0.5),
    alpha = 0.8) +
  geom_point(aes(x = treatment,
                 y = response,
                 fill = lab_or_rw),
    position = position_dodge(width = 0.5),
    alpha = 1,
    size = 3,
    shape = 21,
    color = "black") +
  scale_fill_manual(values = c("Lab" = color_mustard,
                              "RW" = color_green),
    name = expression(bold("Environment")),
    labels = c("Lab" = "Laboratory",
               "RW" = "Rewilded")) +
  scale_color_manual(values = c("Lab" = color_mustard,
                              "RW" = color_green),
    name = expression(bold("Environment")),
    labels = c("Lab" = "Laboratory",
               "RW" = "Rewilded")) +
  scale_y_continuous(name = expression(bold("TNF-" * alpha ~
                                         "(pg mL"^-1" * ")")),
    expand = expansion(mult = 0),
    trans = "log10",
    breaks = 10^seq(-1, 4),
    labels = expression(10^-1, 1, 10^1,
                        10^2, 10^3, 10^4),
    limits = c(0.1, 10000),
    guide = guide_axis_logticks(long = 1.8,
                                mid = 1.3,
                                short = 0.8,
                                expanded = FALSE)) +
  scale_x_discrete(name = expression(bold("Treatment")),
    labels = c("CD28" = "CD28SA",
               "LPS" = "LPS",

```

```

"lgG" = "IgG1")) +
facet_wrap(~ t_muris_dose,
  labeller = as_labeller(c("Zero" = "Zero Dose",
    "High" = "High Dose")))) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
  color = "black",
  linewidth = 0.8),
  strip.text = element_text(size = 10,
    face = "bold"),
  panel.border = element_rect(color = "black",
    fill = NA,
    linewidth = 0.8),
  panel.grid = element_blank(),
  plot.title = element_text(hjust = 0.5, face = "bold"),
  axis.title.x = element_text(face = "bold"),
  axis.title.y = element_text(face = "bold"),
  axis.text = element_text(color = "black"),
  axis.ticks = element_line(color = "black",
    linewidth = 0.3),
  axis.ticks.length = unit(0.1, "cm")) +
ggtitle(expression(bold("EMM of TNF- $\alpha$  * alpha ~ "by Dose"))))
emm_tnf_dose_plot

```




```
plot_saver("emm_tnf_dose_plot.png",
  emm_tnf_dose_plot)
```

EMM TNF-alpha: #3 — Facet = Environment, x = Treatment, Color = Dose

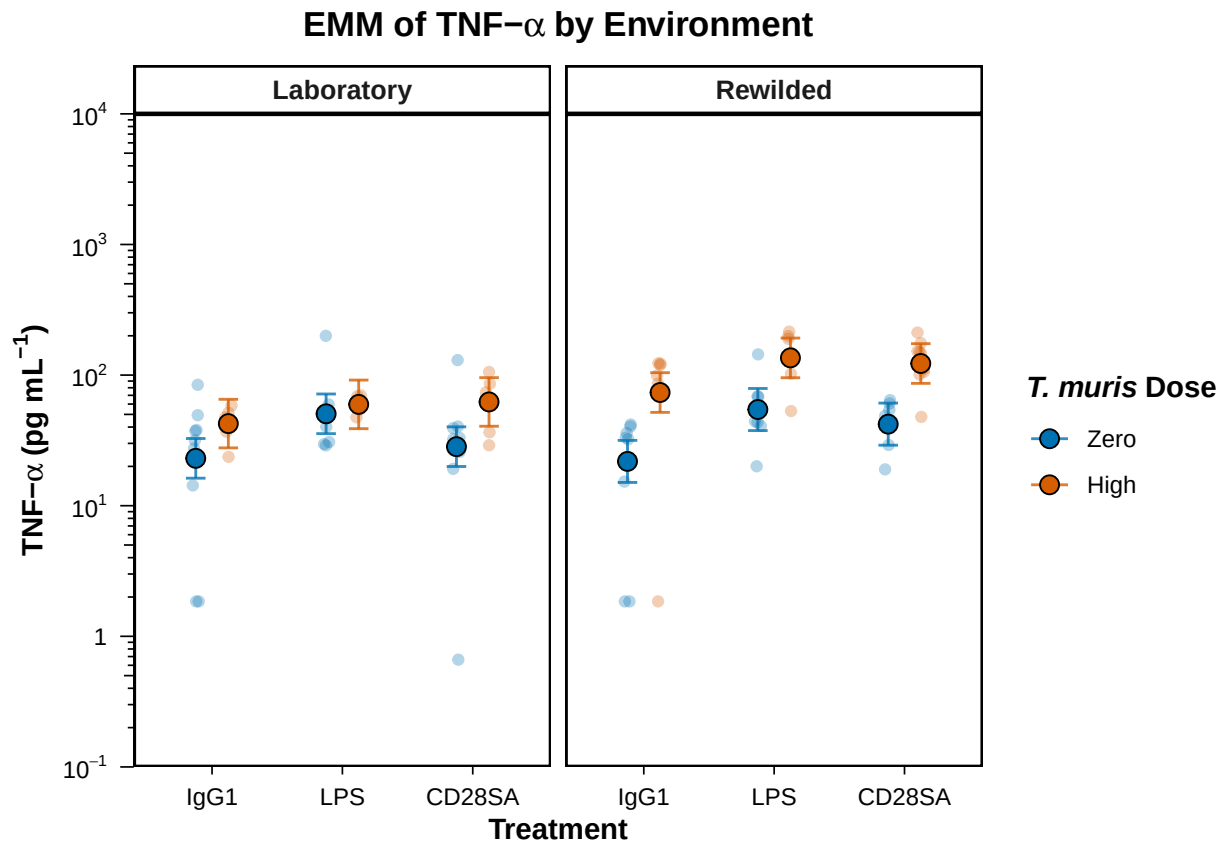
```
emm_tnf_treatment_plot <- ggplot(emm_tnf_df) +
  geom_point(data = tnf_glmer_input %>%
    mutate(treatment = factor(treatment,
                              levels = c("IgG", "LPS", "CD28")),
           lab_or_rw = factor(lab_or_rw, levels = c("Lab", "RW")),
           t_muris_dose = factor(t_muris_dose, levels = c("Zero", "High"))),
    aes(x = treatment,
        y = mean_concentration,
        color = t_muris_dose),
    position = position_jitterdodge(jitter.width = 0.1,
                                    dodge.width = 0.5),
    alpha = 0.3,
    size = 1.5) +
  geom_errorbar(aes(x = treatment,
                   ymin = asymp.LCL,
                   ymax = asymp.UCL,
                   color = t_muris_dose),
               width = 0.3,
               position = position_dodge(width = 0.5),
               alpha = 0.8) +
  geom_point(aes(x = treatment,
                 y = response,
                 fill = t_muris_dose),
             position = position_dodge(width = 0.5),
             alpha = 1,
             size = 3,
             shape = 21,
             color = "black") +
  scale_fill_manual(values = c("Zero" = color_dark_blue,
                              "High" = color_orange),
                   name = expression(bolditalic("T. muris") ~ bold("Dose")) +
  scale_color_manual(values = c("Zero" = color_dark_blue,
                              "High" = color_orange),
                    name = expression(bolditalic("T. muris") ~ bold("Dose")) +
  scale_y_continuous(name = expression(bold("TNF-" * alpha ~
                                         "(pg mL"^-1" * ")")),
                    expand = expansion(mult = 0),
                    trans = "log10",
                    breaks = 10^seq(-1, 4),
                    labels = expression(10^-1, 1, 10^1,
                                         10^2, 10^3, 10^4),
                    limits = c(0.1, 10000),
                    guide = guide_axis_logticks(long = 1.8,
                                                  mid = 1.3,
```

```

                                short = 0.8,
                                expanded = FALSE)) +
scale_x_discrete(name = expression(bold("Treatment")),
  labels = c("CD28" = "CD28SA",
             "LPS" = "LPS",
             "IgG" = "IgG1")) +
facet_wrap(~ lab_or_rw,
  labeller = as_labeller(c("Lab" = "Laboratory",
                           "RW" = "Rewilded")))) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                       color = "black",
                                       linewidth = 0.8),
      strip.text = element_text(size = 10,
                                face = "bold"),
      panel.border = element_rect(color = "black",
                                  fill = NA,
                                  linewidth = 0.8),
      panel.grid = element_blank(),
      plot.title = element_text(hjust = 0.5, face = "bold"),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      axis.text = element_text(color = "black"),
      axis.ticks = element_line(color = "black",
                                linewidth = 0.3),
      axis.ticks.length = unit(0.1, "cm")) +
ggtitle(expression(bold("EMM of TNF- $\alpha$  ~ "by Environment"))))

emm_tnf_treatment_plot

```



```
plot_saver("emm_tnf_treatment_plot.png",
  emm_tnf_treatment_plot)
```

Comparing IFN and TNF levels to see if there is a correlation

```
ifn_tnf_combined <- left_join(ifn_glmr_input,
  tnf_glmr_input,
  by = c("mouse_id",
    "treatment"),
  suffix = c("_ifn",
    "_tnf")) %>%

select("mouse_id",
  "plate_ifn",
  "lab_or_rw_ifn",
  "strain_ifn",
  "t_muris_dose_ifn",
  "cohort_ifn",
  "experiment_ifn",
  "source_ifn",
  "experiment_day_ifn",
  "treatment",
  "mean_concentration_ifn",
  "mean_concentration_tnf") %>%
```

```

rename("plate" = "plate_ifn",
       "lab_or_rw" = "lab_or_rw_ifn",
       "strain" = "strain_ifn",
       "t_muris_dose" = "t_muris_dose_ifn",
       "cohort" = "cohort_ifn",
       "experiment" = "experiment_ifn",
       "source" = "source_ifn",
       "experiment_day" = "experiment_day_ifn") %>%
mutate(treatment = factor(treatment,
                          levels = c("CD28", "LPS", "IgG")))

```

We used log scale here to make sure that the whole graph fits.

```

ifn_tnf_combined_plot <- ggplot(ifn_tnf_combined %>%
                                mutate(treatment = factor(treatment,
                                                            levels = c("IgG",
                                                                    "LPS",
                                                                    "CD28"))),
                                aes(x = mean_concentration_ifn,
                                    y = mean_concentration_tnf,
                                    fill = lab_or_rw,
                                    color = lab_or_rw,
                                    group = lab_or_rw)) +
  geom_point(aes(shape = t_muris_dose),
             size = 2,
             alpha = 0.8) +
  geom_smooth(method = "lm",
              se = TRUE,
              alpha = 0.1,
              linewidth = 0.8) +
  facet_grid(~ treatment,
             labeller = as_labeller(c("CD28" = "CD28SA",
                                       "LPS" = "LPS",
                                       "IgG" = "IgG1")))) +
  scale_fill_manual(values = c("RW" = color_green,
                              "Lab" = color_mustard),
                   name = expression(bold("Environment")),
                   labels = c("Lab" = "Laboratory",
                              "RW" = "Rewilded")) +
  scale_color_manual(values = c("RW" = color_green,
                              "Lab" = color_mustard),
                    name = expression(bold("Environment")),
                    labels = c("Lab" = "Laboratory",
                              "RW" = "Rewilded")) +
  scale_shape_manual(values = c("Zero" = 1,
                              "High" = 3),
                    name = expression(bolditalic("T. muris") ~ bold("Dose")) +
  scale_x_continuous(name = expression(bold("IFN- $\gamma$ " ~ "(pg mL-1"))),
                    trans = "log10",
                    breaks = 10^seq(-1, 3),
                    labels = expression(10-1,

```

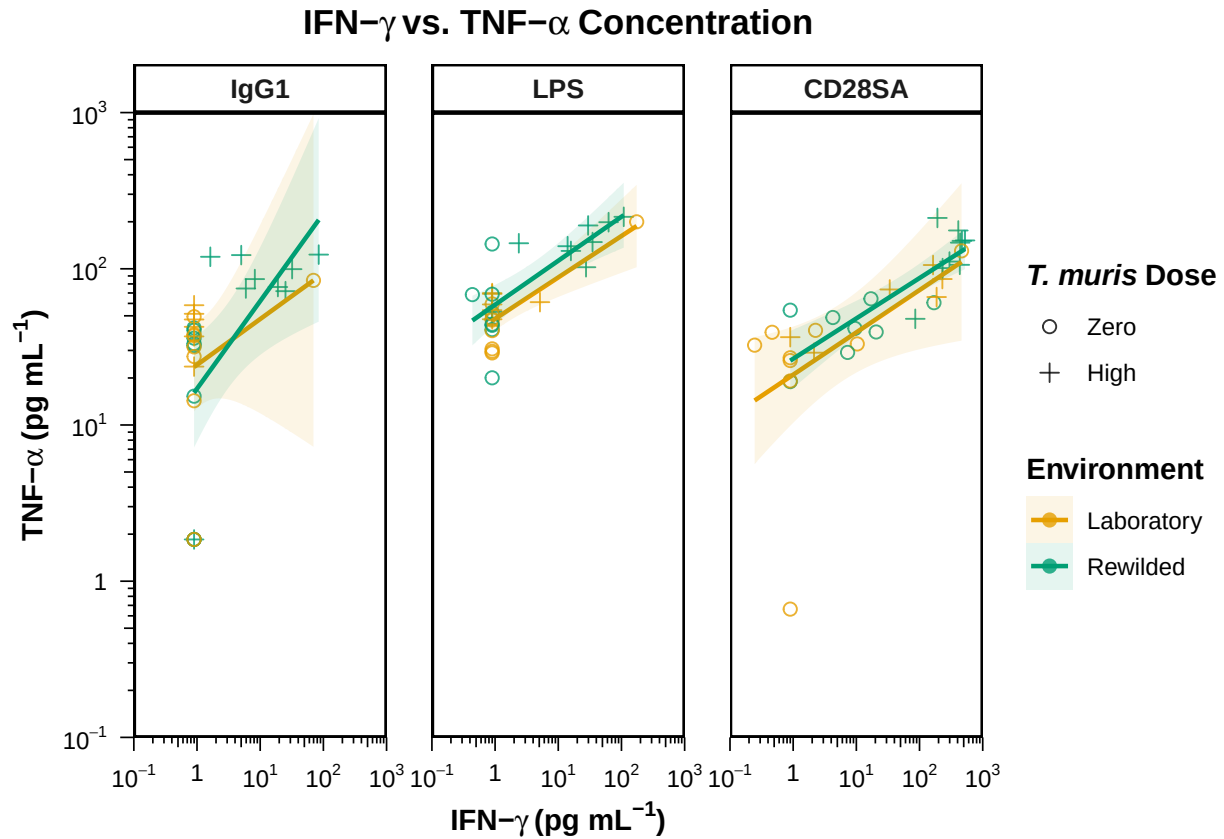
```

1,
10^1,
10^2,
10^3),
limits = c(0.1, 1000),
expand = expansion(mult = 0),
guide = guide_axis_logticks(long = 1.8,
                             mid = 1.3,
                             short = 0.8,
                             expanded = FALSE)) +
scale_y_continuous(name = expression(bold("TNF- $\alpha$ "(pg mL)-1")),
                    trans = "log10",
                    breaks = 10^seq(-1, 3),
                    labels = expression(10-1,
                                         1,
                                         101,
                                         102,
                                         103),
                    limits = c(0.1, 1000),
                    expand = expansion(mult = 0),
                    guide = guide_axis_logticks(long = 1.8,
                                                  mid = 1.3,
                                                  short = 0.8,
                                                  expanded = FALSE)) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                       color = "black",
                                       linewidth = 0.8),
      strip.text = element_text(size = 10,
                                face = "bold"),
      panel.border = element_rect(color = "black",
                                  fill = NA,
                                  linewidth = 0.8),
      panel.grid = element_blank(),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      axis.text = element_text(color = "black"),
      axis.ticks = element_line(color = "black",
                                linewidth = 0.3),
      axis.ticks.length = unit(0.1, "cm"),
      plot.title = element_text(hjust = 0.5,
                                face = "bold"),
      panel.spacing = unit(0.6, "cm")) +
ggtitle(expression(bold("IFN- $\gamma$ "vs. TNF- $\alpha$ "Concentration")))

ifn_tnf_combined_plot

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
plot_saver("ifn_tnf_combined_plot.png",
  ifn_tnf_combined_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
# Highest IFN
ifn_tnf_combined %>%
  slice_max(mean_concentration_ifn, n = 10) %>%
  select(mouse_id, mean_concentration_ifn)
```

```
## # A tibble: 10 x 2
##   mouse_id mean_concentration_ifn
##   <dbl>         <dbl>
## 1     1368         525.
## 2     1406         468.
## 3     1407         463
## 4     1363         440.
## 5     1396         434.
## 6     1359         411.
## 7     1373         298.
## 8     1378         232.
## 9     1370         230.
## 10    1401         192.
```

```
# Highest TNF
ifn_tnf_combined %>%
  slice_max(mean_concentration_tnf, n = 10) %>%
  select(mouse_id, mean_concentration_tnf)
```

```
## # A tibble: 10 x 2
##   mouse_id mean_concentration_tnf
##   <dbl>         <dbl>
## 1     1406           215.
## 2     1401           212.
## 3     1407           200.
## 4     1363           199.
## 5     1401           189.
## 6     1359           176.
## 7     1368           152.
## 8     1406           151.
## 9     1396           148.
## 10    1363           147.
```